

Weather, Monitors, or Policy?

A meteorologically normalised, quasi-experimental evaluation of India's National Clean Air Programme, 2015–2025

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Abstract

Public progress reports compare a city's raw pollution with a baseline year, and in the cities where I could check this, that comparison overstates the improvement. In the 18 NCAP cities that had a PM2.5 monitor valid every year from 2018 to 2025 (16 of them with only one such monitor), the reported fall of 23.6% becomes 14.1% once I remove modelled weather and the change in which monitors exist. Most of the difference comes from the changing set of monitors: the stations added since 2018 read cleaner than the ones already there (H4 supported: +9.5 percentage points, 95% CI +4.8 to +14.5). Weather alone moves a typical city's annual PM2.5 by a median 5.1% from one year to the next. For the policy question, I compared the 113 urban centres containing 121 of NCAP's 131 cities with 923 comparable non-NCAP centres on satellite PM2.5. Satellite PM2.5 fell in both groups after 2018: by 16.8% in NCAP units and by 24.3% in the comparison pool. NCAP units' satellite PM2.5 did not fall relative to comparable units; the estimates point to a relative rise of about 3–5% (primary estimate +3.6%, 95% CI +2.4% to +4.8%). But the pre-trend test I registered in advance failed ($p = 0.0008$), so H1 is not identified by this design, and I do not attribute this or any other number to NCAP. Raw aerosol optical depth, which no ground monitor calibrates, shows a smaller relative change than the satellite PM2.5 product on the same units in every specification, for reasons this design cannot separate. The gap seen in satellite PM2.5 between units that gained a monitor and units that did not is absent from raw aerosol optical depth, which is consistent with, but does not prove, the satellite product absorbing the new monitors; an exploratory comparison, in which the two signals diverge most where no monitor was added, weakens it as evidence for leakage specifically. City-level differences are too uncertain to rank cities. The registered tests for a smaller change in the Indo-Gangetic Plain (H5) and for dust control (H3) came out “not met” and “inconclusive”.

I registered the analysis plan on OSF before comparing NCAP with non-NCAP cities after 2018: <https://osf.io/jksne/>. Code, data provenance and every decision log are at <https://github.com/breenu/ncap-evaluation>, and a city-by-city explorer is at <https://breenu.github.io/ncap-evaluation/>.

What this report does and does not claim

- **H1 is not identified by this design.** My registered pre-trend test failed, so no number in this report is an effect of NCAP. Wherever a satellite estimate appears, the verdict appears with it: NCAP units' satellite PM2.5 did not fall relative to comparable units; the estimates point to a relative rise of about 3–5%. H1 is not identified by this design, so this is not an effect of NCAP.
- **No city is ranked.** City-level estimates are too uncertain to order, and every named list is alphabetical.
- **Ground (Layer B) results are secondary.** They rest on one pre-NCAP year and, mostly, on single monitors; each carries its scope.
- **Raw aerosol optical depth is read for direction only**, because it is not surface PM2.5.
- **No claim is made about any individual weather variable.**

1 Introduction

1.1 NCAP and how its progress is reported

India's National Clean Air Programme (NCAP) began in January 2019. At its widest it covered 131 cities that had breached the national standard for particulate matter or have a million or more people; the list current in 2026 has 130. Its targets are for PM10: reductions of 20%, 30%, 40% against a baseline, the last by the financial year that ended in March 2026. Cities receive central money through two channels, the Ministry's NCAP budget (82 cities) and performance-linked grants from the XV Finance Commission to million-plus cities (49).

Official and independent trackers judge progress by comparing a city's measured concentration in a recent year with its baseline. That comparison mixes three things.

- **Weather.** A windy year or an early monsoon lowers pollution without any change in emissions.
- **Which monitors exist.** NCAP set out to expand the continuous monitoring network (NCAP report, January 2019, section 8.1), and its funds support city action plans that include expanding it (Lok Sabha answer AU5104, April 2022). The network grew rapidly after NCAP began. If new monitors are placed in cleaner neighbourhoods, a city's average falls even if nobody's air changed. The programme changed the thermometer used to judge it.
- **Policy.** Only what remains after the first two could reflect NCAP, and only if the cities that were listed can be compared with cities that were not.

This report separates the three.

1.2 Research questions and hypotheses

- **RQ1, measurement:** how much of a city's reported change comes from data-quality problems and from changes in which stations exist?
- **RQ2, weather:** how much of the year-to-year change comes from meteorology?
- **RQ3, policy:** did PM2.5 change differently in NCAP cities than in comparable non-NCAP cities? This is the confirmatory question.
- **RQ4, heterogeneity and mechanism:** does any change vary with region and funding, and does the PM10/PM2.5 pattern fit dust control?

Table 1. The registered hypotheses and their results. Layer A is satellite PM2.5 over urban centres; Layer B is ground monitors.

status	layer	hypothesis (registered)	result
H1 confirmatory	A	NCAP enrolment reduced annual population-weighted PM2.5 relative to comparable centres	not identified by this design (pre-trend Wald p = 0.0008)
H2 confirmatory, only if H1 is supported	A	the reduction is larger in winter	not tested (H1 not supported); exploratory difference +0.1%
H3 secondary	B	PM10 fell more than PM2.5 and the PM2.5/PM10 ratio rose (dust control)	inconclusive
H4 secondary, descriptive	B	reported improvements exceed weather- and composition-corrected ones	supported: PM2.5 +9.5 pp, PM10 +6.0 pp
H5 secondary	A	smaller (less negative) changes in Indo-Gangetic Plain centres	registered rule not met

1.3 Pre-registration

I wrote the analysis plan before comparing NCAP with non-NCAP cities after 2018 and registered it on OSF on 27 September 2026 (<https://osf.io/jksne/>; the registered file is the plan at commit [6e24eca](#)). The pipeline enforced this in code: every step that estimates a post-2019 comparison checks a gate file and refuses to run while it is shut, and the gate was opened in its own commit ([0d9aa42](#)) citing the registered plan. Before registering I had seen pre-2019 data, post-2019 series for single cities and regions, and data-quality diagnostics, but no comparison of NCAP with non-NCAP units after 2018. On 1 October 2026 I posted a clarification of how I read some registered rules whose sign was ambiguous. Every departure from the plan is listed, dated and justified in Appendix A.

1.4 What is new here

- A station-level audit of India’s continuous monitoring network, with a reliability score for every station-year.
- A systematic measure of network composition: how much the changing set of monitors moves reported city trends.
- Weather normalisation combined with a long satellite pre-period and a comparison group of non-NCAP cities.
- A falsifiable mechanism test (PM10 against PM2.5), and a pre-registered design in which a failed identification check is reported as such.

2 Data

Table 2. Data sources. Every raw file is stored unchanged with a manifest of its source, download date and checksum.

Source	What I use	Version and coverage	Licence
CPCB continuous stations, via the india-cpcb-aqi mirror (Vonter)	15-minute PM2.5 and PM10	Releases of January 2026; 2015–2025	ODbL 1.0; some contents © CPCB
OpenAQ	Station coordinates; an independent copy of CPCB data for cross-checks; January–March 2026 (not used in primary results)	S3 archive and v3 API, snapshot of September 2026	Attribution to OpenAQ and the original providers
ACAG satellite PM2.5 (Washington University in St. Louis)	Annual and monthly surface PM2.5	V5.GL.06 (primary), V6.GL.03, V6.GL.02.04; 2005–2024	CC BY 4.0; cite the product papers
ERA5 (Copernicus Climate Change Service)	Hourly weather at every station’s grid cell; monthly means over the satellite units	2015 to March 2026 (hourly); 2005–2024 (monthly)	CC BY 4.0
GHSL Urban Centre Database and GHS-POP (European Commission, JRC)	Urban-centre polygons; population weights	UCDB R2024A V1-2; GHS-POP R2023A, epoch 2020	CC BY 4.0
DataMeet India boundaries	State boundaries and regions; maps	Repository commit b3fbbde	CC BY 4.0
GeoNames	Points for NCAP towns without a GHSL centre, and a few station localities	Extract of 26 September 2026	CC BY 4.0
Natural Earth	Coastline, for the coastal region	1:10m coastline	Public domain
NASA FIRMS (VIIRS S-NPP)	Active fires, for one robustness check	2012–2024	NASA open data, with acknowledgement
MODIS MAIAC aerosol optical depth (NASA LP DAAC), via Google Earth Engine	Raw AOD, reduced to the same polygons, for the calibration check	MCD19A2 Collection 6.1; 2010–2024	No restriction on use or redistribution
NCAP documents (CPCB, MoEFCC, XV Finance Commission, Lok Sabha answers)	City lists and their dates; funding channel; allocations	Fourteen documents, 2017–2026, each row traced to its page	Government of India publications; facts extracted

2.1 Ground monitors

The ground data are CPCB’s continuous ambient air quality stations, 2015–2025. CPCB’s own data repository stopped answering during this project, and OpenAQ holds almost no CPCB data for 2023 and 2024, so I used a public mirror of CPCB’s repository (Vonter, [india-cpcb-aqi](#), ODbL). I did not trust it blindly. Every file matches the checksum the mirror publishes, and its values match OpenAQ’s independent copy of the same stations value for value where both exist (2019–2021 and 2025). The mirror stores Indian clock times labelled as UTC; I confirmed the correct offset with three independent tests, including the daily peak of solar radiation at each station. January to March 2026 exists only in OpenAQ’s unvalidated real-time feed, so it is not used in any primary result.

The network changed under NCAP (Figure 2). Of the 565 stations, 139 first reported by 2018 and 426 from 2019 onward. The ground record before NCAP is therefore thin: PM10 has 9 valid station-years in 2017 and 66 in 2018.

Figure 2. Did the measuring instrument change under the programme?

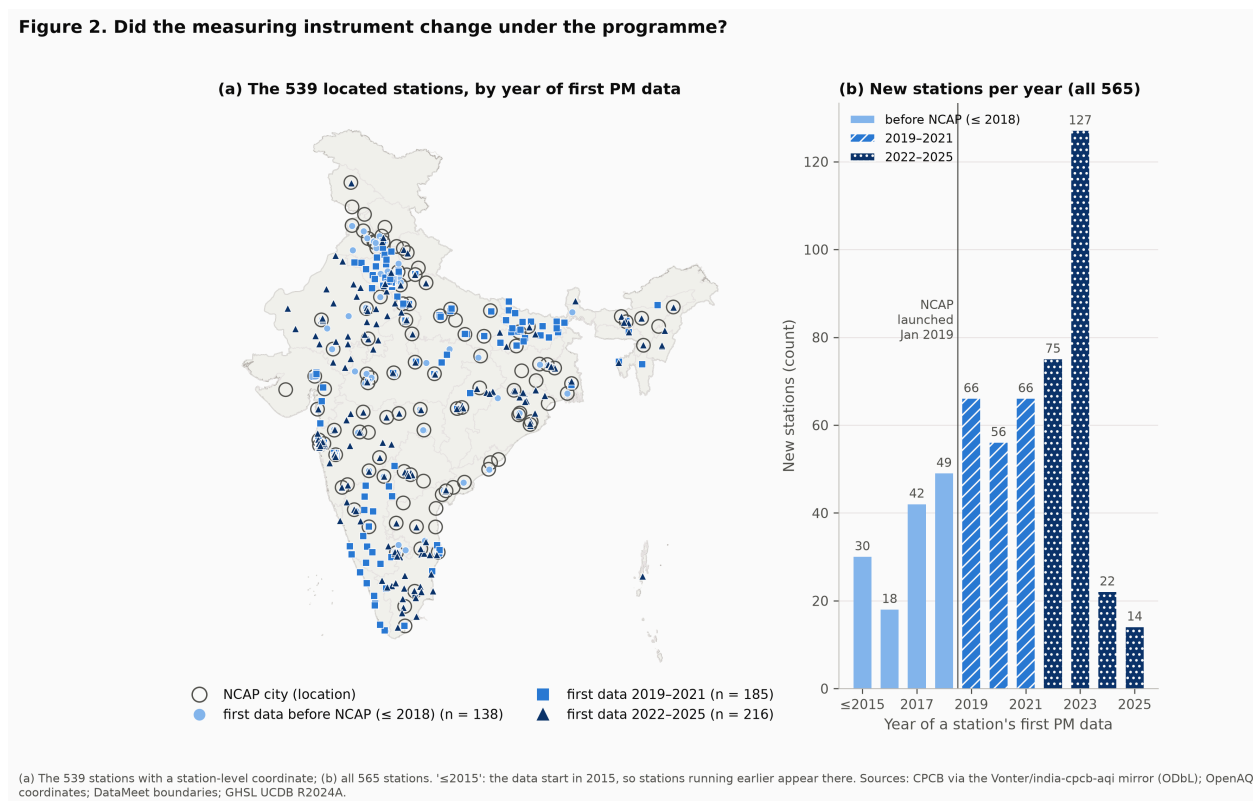


Figure 2. (a) The 539 CPCB continuous monitoring stations with a station-level coordinate, by the period of their first PM data: 138 before NCAP (to 2018), 185 in 2019–2021 and 216 in 2022–2025; open circles mark NCAP cities. (b) New stations per year, all 565 stations: 139 first reported in 2015–2018 and 426 from 2019 onward, peaking at 127 in 2023. Most of the network that measures NCAP cities today did not exist at the 2018 baseline, which is why reported city averages mix air-quality change with changes in where monitors stand.

2.2 Satellite PM2.5

The primary outcome is the Atmospheric Composition Analysis Group's satellite-derived surface PM2.5, version V5.GL.06, which is documented and covers 1998–2024. V6.GL.03 (a different algorithm) and V6.GL.02.04 (an earlier calibration vintage) are sensitivity checks. The product is calibrated against ground monitors, so new monitors could leak into it; I test this in Section 4.4.

2.3 Units, treatment and the comparison group

The unit is a GHSL urban centre (R2024A). A city's satellite value is the population-weighted mean over its polygon, for every unit alike, so that it measures what residents breathe and needs no arbitrary clipping. NCAP cities that share a polygon form one unit (Delhi, Faridabad, Ghaziabad and Noida share one).

- **Treated:** 113 units holding 121 of the 131 NCAP cities. The 10 NCAP towns with no GHSL centre enter only a sensitivity check.
- **Treatment date:** the first official list containing the city, reconstructed from dated versions of CPCB's list and NCAP documents, every row traced to its source page. Listed by 30 June counts from that calendar year. This gives cohorts of 89 units (2019), 15 (2020) and 9 (2021). The year of first central funding is a sensitivity check only, because XV Finance Commission grants reward cities that improved, so the timing of money is partly an outcome.
- **Comparison pool:** non-NCAP centres with at least 100,000 people in 2015, a pre-treatment year: 923 units.

NCAP units are much larger than the comparison pool (standardised difference in log population 1.44) and less often in the Indo-Gangetic Plain (28% against 43%), but their mean 2010–2018 PM2.5 is similar (51.7 against 56.4 $\mu\text{g}/\text{m}^3$).

2.4 Weather, fires and aerosol optical depth

Weather comes from ERA5: hourly series at the grid cell of every station, and monthly means for the satellite units. Fires come from NASA FIRMS (VIIRS, from 2012). For the calibration check I reduced raw MODIS MAIAC aerosol optical depth (AOD), a satellite signal that uses no ground monitor, to the same polygons on Google Earth Engine.

3 Methods

3.1 Audit and validity rules (RQ1)

Every rule is a pure function with a unit test on synthetic data.

- **Impossible values and ceilings.** Values at or below zero are removed. Instrument ceilings (values pinned at the top of an analyser's range) are detected from the network-wide histogram rather than typed in: a value of at least $500 \mu\text{g}/\text{m}^3$ that is far more frequent than its neighbours at many stations.
- **Flatlines:** identical hourly means for 4 or more hours.
- **PM2.5 above PM10** at the same station-hour, beyond the larger of $5 \mu\text{g}/\text{m}^3$ and 10% of PM10.
- **Level shifts:** changepoints in each station's weekly departure from its neighbours, with the penalty calibrated so that at most 5% of shuffled series with no shift show one; only steps of at least 25% count.
- **Spatial consistency:** each station-year against its neighbours within 25 km and against its satellite cell.
- **Completeness:** a valid day needs 75% of its hours and a valid year 75% of its days; 60% and 90% are sensitivity checks.
- **Stuck analysers** (a deviation, A2): a station-year is excluded if its 15-day variability stays far below its neighbours' for at least 30 days.
- **Reliability score:** completeness $\times$ the share of unflagged data, multiplied by 0.8 for each failed consistency check. It describes data quality and excludes nothing.
- **Missingness:** linear probability models of a station-day being missing, by season and by the pollution level the neighbours recorded that day.

3.2 Deweathering (RQ2)

For each station and pollutant, a model predicts daily log PM from that day's weather, the calendar and a slow trend that stands for emissions (Grange and Carslaw's method). The model then predicts each day 500 more times with weather drawn from other days and averages, giving the concentration expected under typical weather. Only weather is resampled; the trend stays.

- **Two model families:** a GAM (R `mgcv`) whose trend has knots exactly one year apart, so that it cannot follow seasons or single weather episodes (deviation A3), and LightGBM.
- **Resampling scheme:** weather is drawn from ERA5 days within ± 15 days of the same date in any year 2015–2025 (deviation A1). Grange and Carslaw's all-year scheme is a sensitivity check.
- **Choosing the family:** the registered rule picks the family with the better median out-of-sample R^2 under blocked, forward-chaining cross-validation by year. Random splits would be dishonest because pollution episodes last several days.
- **Guards:** I report how often the resampled weather leaves a station's training range, and flag station-years whose deweathered mean is outside $0.67\text{--}1.5\times$ the raw mean.

3.3 Network composition and H4 (RQ1)

For every city I build three annual trends: all stations as reported; a balanced panel of the stations inside the city's polygon that were valid in 2018 and every year to 2025; and satellite PM2.5 over the same polygon. The reported change from 2018 to 2025 then splits exactly into four parts:

reported = unmodelled change + modelled weather + network composition + corrected change.

The corrected change is the deweathered balanced panel. H4 is the reported fall minus the corrected fall, so a positive value means the reported number overstates the improvement. Uncertainty across cities comes from a cluster bootstrap over cities (1,000 resamples). H4 describes NCAP cities with a continuous station only; it is descriptive, not causal.

3.4 Layer A: the satellite comparison (RQ3, confirmatory)

- **Primary estimator:** synthetic difference-in-differences (SDID), estimated per listing cohort against never-treated centres, 2010–2024 with 2020 dropped, and combined by the number of treated units. The standard error comes from 500 joint placebo replications in which random control sets of the cohorts' sizes are given the cohorts' adoption years.
- **Event study:** the Sun and Abraham estimator with unit and region $\times$ year effects and ERA5 covariates, relative years -9 to $+5$; Callaway and Sant'Anna's estimator as the staggered-adoption check; Rambachan and Roth's HonestDiD bounds on the pre-trends.

- **Decision rules (registered):** H1 is supported only if (a) the SDID estimate is a reduction with a 95% interval below zero, (b) the pre-period event-study coefficients are jointly zero (Wald $p > 0.10$), (c) a placebo listing in 2016 shows nothing, and (d) the sign holds in Callaway and Sant’Anna, the area-weighted outcome and V6.GL.03. **If (b) or (c) fails, the verdict is “not identified by this design”, whatever the estimate.** If the interval included zero, an equivalence test against $\pm 5\%$ would apply.
- **Minimum detectable effect:** estimated before the gate from placebo listings in 2014 and 2015 on pre-2019 data only: a 1.2% fall, read as an optimistic lower bound.
- **Calibration leakage:** the primary design is re-estimated separately for treated units that gained a monitor inside their polygon in 2019–2024 and for those that did not. A pre-stated rule flags a warning if the difference points to leakage.
- **Raw AOD:** the same design on log AOD, read only for direction against satellite PM2.5 on exactly the same units, with categories written before any AOD value existed.

3.5 Layer B: the ground comparison, and triangulation

Layer B uses the deweathered balanced panels of H4. The interrupted time series (ITS) is each city’s mean log level after listing minus before, which with one pre-year is a before–after contrast. The ground difference-in-differences (DiD) compares NCAP stations with stations in control-pool cities, per cohort. Both are secondary. The two layers are never averaged. Layer A restricted to the cities with ground panels is set against each Layer B estimator and classified in a fixed order: uninformative, conflict, consistent, different magnitude. Any category other than “consistent” triggers a registered four-step investigation (network composition, satellite calibration, spatial coverage, deweathering), reported whether or not it closes the gap.

3.6 Heterogeneity and mechanism (RQ4)

- **City-level estimates:** one SDID per treated unit against all controls. Its standard error is the spread of the same estimate when each control in turn is treated as the listed unit.
- **Hierarchical model:** a Bayesian measurement-error model (PyMC; 4 chains of 2,000 draws) pools the city estimates with the five registered moderators: baseline PM2.5, the Indo-Gangetic Plain, log population, coastal location and funding channel. H5 is decided by the IGP coefficient.
- **H3 (dust control)** needs three conditions: the PM2.5/PM10 ratio rises; PM10 falls, and more than PM2.5, in at least two of three specifications for both ground estimators; and the satellite and ground layers do not conflict.
- **Funding dose (exploratory):** city-level changes against XV Finance Commission allocations per person. Allocations rather than releases are used, because releases reward improvement.

Table 3. The registered decision rules for H1 and their values. The full list of rules as I read them is in DEC-135 and Appendix A (group C).

registered rule (as read in DEC-135)	value	met
(a) primary SDID estimate < 0 , 95% CI entirely below 0	+3.6% (+2.4% to +4.8%)	no
(b) event-study pre-period coefficients jointly zero (Wald $p > 0.10$)	$\chi^2 = 26.8$, 8 df, $p = 0.0008$	no
(c) 2016 placebo-in-time 95% CI includes 0	−0.1% (−0.9% to +0.7%)	yes
(d) negative in Callaway & Sant’Anna, area-weighted and V6.GL.03	+4.6%; +3.5%; +3.2%	no

4 Results

4.1 The argument in one figure

Figure 1 summarises the report. Panel (a) is RQ1 and RQ2: the reported fall in ground PM2.5 over the H4 cities, and what remains after removing what the model cannot reproduce, modelled weather and the change in which monitors exist. Panel (b) is RQ3, on its own axis because it is a different quantity: how these cities’ satellite PM2.5 moved against comparable centres after listing. It is not identified as an effect of NCAP. Panel (c) repeats panel (a) for four cities chosen by a station-count rule that I wrote after seeing every city’s result (Appendix A, D5); every city is in Figure S3.

Figure 1. How much of NCAP cities' reported fall in PM2.5 survives weather and network correction, and how did their satellite PM2.5 move relative to comparison cities?

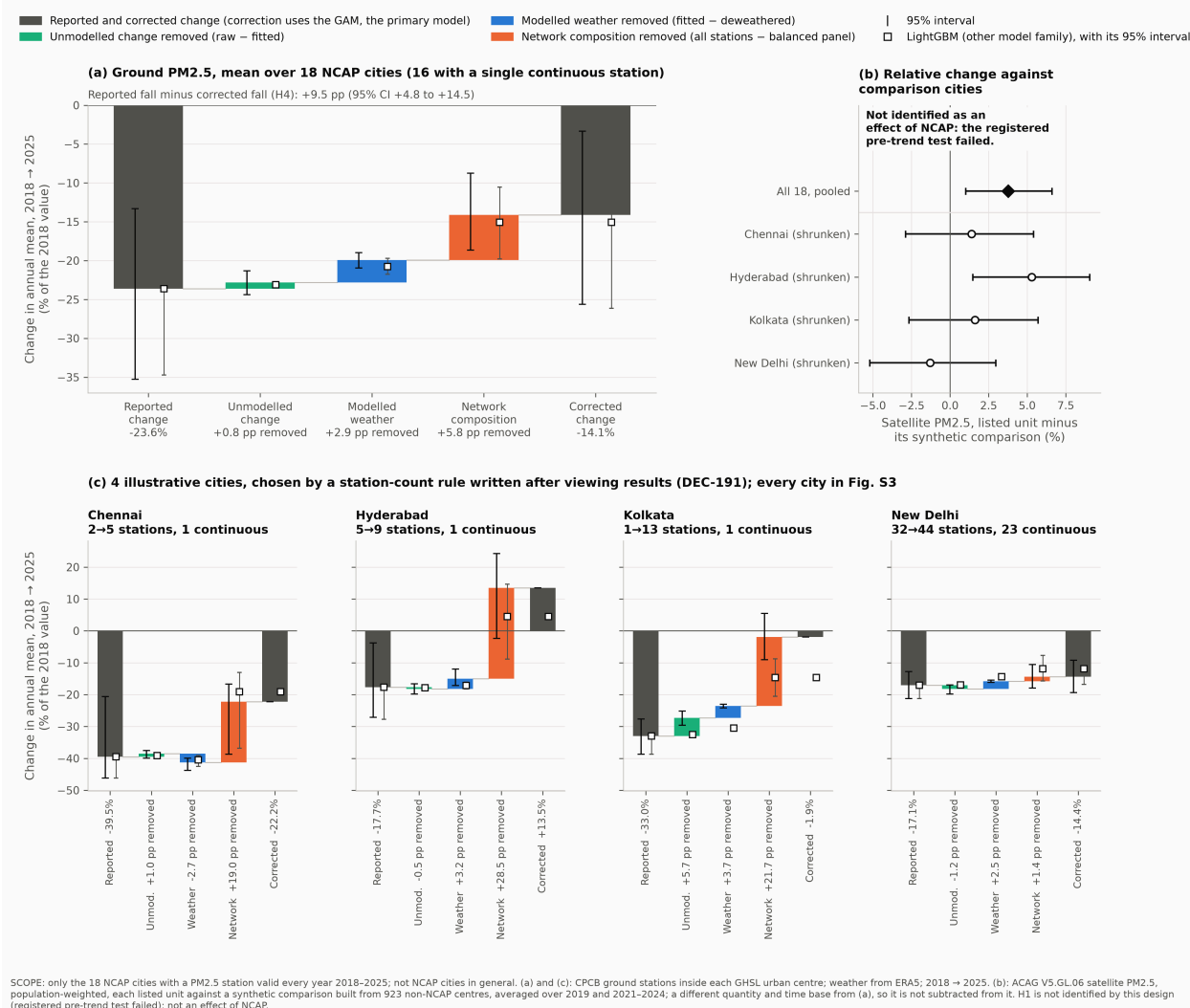


Figure 1. (a) Mean change in annual ground PM2.5 from 2018 to 2025 over the 18 NCAP cities with a station valid every year (16 of them with only one such station), decomposed in the proposal's order. Reported change (all stations as reported, after the audit's cleaning) -23.6%; removing the part the deweathering model does not reproduce (+0.8 pp removed), modelled weather (+2.9 pp removed) and the change in which stations exist (+5.8 pp removed) leaves a weather- and composition-corrected change of -14.1% (balanced panel, deweathered). Bars: GAM (primary); squares: LightGBM; intervals: 95% cluster bootstrap over cities. H4 (reported fall minus corrected fall) = +9.5 pp (95% CI +4.8 to +14.5). (b) Satellite PM2.5 of the listed units relative to their synthetic comparison units after listing: the 18 cities' units pooled (Phase 7, Layer A restricted) +3.8% (95% CI +1.0 to +6.6), and the shrunk city-level relative change of each illustrative city (Phase 8; 95% credible interval): Chennai +1.4% (-2.9 to +5.4); Hyderabad +5.3% (+1.5 to +9.0); Kolkata +1.6% (-2.7 to +5.7); New Delhi -1.3% (-5.2 to +3.0). This is a different quantity on a different time base from (a), so it is drawn on its own axis and not subtracted. Each removed part is labelled with the step it makes, so its sign is the direction of its bar. (c) The same decomposition for Chennai, Hyderabad, Kolkata, New Delhi, chosen by a rule on station counts that was written after viewing the per-city results (DEC-191); every city is in Figure S3. Intervals are station bootstraps where a city has more than one station in a stratum, otherwise none can be estimated and the GAM–LightGBM gap shows the model uncertainty. H1 is not identified by this design (registered pre-trend test failed): not an effect of NCAP.

4.2 RQ1: measurement

The audit. Ceiling values were found at 842, 887, 985, 995, 999.99, 1000 $\mu\text{g}/\text{m}^3$ for PM2.5 and 985, 999.99, 1000 $\mu\text{g}/\text{m}^3$ for PM10; together with impossible values they remove 0.09% of quarter-hour values. Flatlined hours are 0.3% to 0.9% of hours a year from 2017. PM2.5 above PM10 affects 0.4% to 1.6% of hours before 2019 but at most 0.09% from 2020, a sign that CPCB's own validation tightened. I found 653 level shifts at 316 stations; the audit cannot tell a recalibration from a real local change beside a monitor. Most stations track their neighbours closely (median correlation 0.91). The stuck-analyser rule removes 77 of 4,129 valid station-years at 33 stations. Figure 8 shows the reliability score of every station-year.

Missing data are not hiding the dirty days. A station-day is 6.0 percentage points less likely to be missing in winter than in the monsoon, and 4.6 points less likely on the most polluted third of days, judged by the neighbours. So naive annual means are biased slightly upward, not downward: a median +0.7% for PM2.5.

Figure 8. How trustworthy is the PM2.5 network over time?

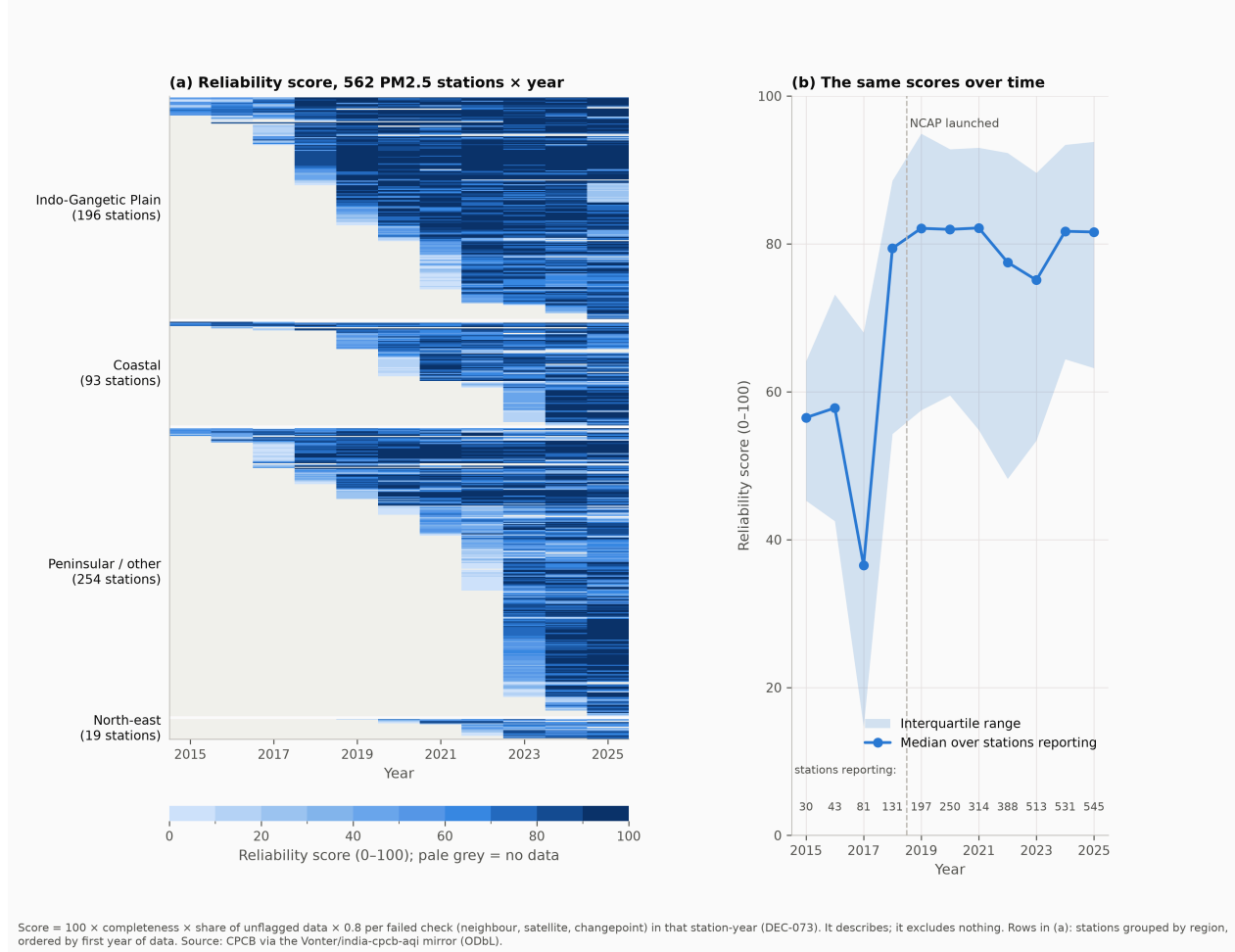


Figure 8. (a) The reliability score (0–100) of every PM2.5 station in every year it reported, 562 stations, grouped by region and ordered by first year of data; pale grey cells have no data. (b) The median score over the stations reporting each year, with its interquartile range: 79 over 131 stations in 2018 and 82 over 545 in 2025. The score is 100 × completeness × share of unflagged data, times 0.8 for each failed consistency check (neighbours, satellite, changepoints). It describes data quality; no data are excluded by it.

Network composition moves reported trends. Across the 23 cities with a 2018 PM2.5 panel, NCAP or not, the changing set of stations makes the average city look 3.7 percentage points better by 2025 than its continuous stations (95% CI –7.2 to –0.4). The gap is near zero until 2021 (–0.5) and appears from 2022 (–1.8, then –4.1 in 2023), when the network grew fastest; by 2025, 13 of these cities measured with a different set of stations than their panel. For PM10 the bias is +0.9 points (–1.1 to +3.3).

New stations read cleaner, and not because of weather. In the same city and year, PM2.5 at newly opened stations is 6.3% below the existing stations (95% CI –11.7 to –0.3), and 6.8% below after deweathering (–12.4 to –0.9); the paired difference is –0.6 points (–2.2 to +1.1). Weather is shared within a city-year, so it cancels: the gap is about where new monitors stand. PM10 shows no clear gap (–1.5%, –6.9 to +5.3).

Ground and satellite agree on average, not city by city. From 2018 to 2024, the balanced-panel change minus the satellite change averages +3.0 points over 23 cities (–5.0 to +10.6), and the two changes correlate at 0.68 across cities. Single monitors often fell much more or much less than the satellite says about their whole polygon.

H4: supported, for both pollutants. In the 18 NCAP cities with a PM2.5 station valid every year from 2018 to 2025 (16 with a single such station), the reported change of –23.6% becomes –14.1%: about 40% of the reported fall does not survive correction. Composition is the larger part for PM2.5 (+5.8 points of +9.5); for PM10 it is modelled weather (+5.4 of +6.0). H4 is supported in 31 of 31 versions for PM2.5 and 29 of 31 for PM10. The two PM10 versions where it fails are: LightGBM, seasonal, primary rule, baseline 2019: +1.7 pp (–0.1 to +3.5); none (raw), no deweathering,

primary rule: +0.1 pp (−2.1 to +2.4). The second measures composition alone. The weather part compares two single years, 2018 and 2025, because that is how H4 was registered, and 2025’s weather was favourable.

Table 4. H4 for PM2.5 and PM10, in the primary version, with LightGBM, and as registered with no deviations. Changes are 2018 to 2025 in %; parts are percentage points of the reported change; H4 = reported fall minus corrected fall. Scope: NCAP cities with a station valid every year 2018–2025, mostly single stations; descriptive, not causal.

version	pol- lu- tant	cities (single- station)	reported change	corrected change	unmod- elled (pp)	modelled weather (pp)	composi- tion (pp)	H4, pp (95% CI)	ver- dict
Primary (GAM)	PM2.5	18 (16)	−23.6%	−14.1%	+0.8	+2.9	+5.8	+9.5 (+4.8 to +14.5)	supported
Primary (GAM)	PM10	13 (11)	−9.6%	−3.6%	−0.5	+5.4	+1.0	+6.0 (+1.8 to +10.0)	supported
LightGBM	PM2.5	18 (16)	−23.6%	−15.1%	+0.5	+2.3	+5.7	+8.6 (+4.7 to +12.6)	supported
LightGBM	PM10	13 (11)	−9.6%	−4.3%	+0.2	+4.5	+0.5	+5.3 (+3.1 to +7.7)	supported
As registered, no de- viations	PM2.5	18 (16)	−23.7%	−13.5%	+1.5	+27.7	−19.0	+10.3 (+5.3 to +14.8)	supported
As registered, no de- viations	PM10	13 (11)	−9.6%	−3.0%	+0.9	+10.9	−5.3	+6.5 (+3.9 to +9.5)	supported

In the “as registered, no deviations” rows, the split into parts is unstable (for PM2.5, modelled weather +27.7 pp and composition −19.0 pp). The original GAM trend under all-year resampling extrapolates to implausible values in a few all-station station-years, the problem that motivated deviations A1 and A3. Only its H4 total, which uses the balanced panel alone, is comparable with the other rows.

4.3 RQ2: weather

The registered rule chose the GAM: its median out-of-sample R^2 is 0.475 against LightGBM’s 0.432, and it predicts better in 756 of 946 series (1,054 series in all). Under the other cross-validation convention the GAM still wins (0.469 against 0.450).

Weather moves a city’s annual PM2.5 by a median 5.1% a year. Across 171 cities, the median year-on-year change in annual PM2.5 is 10.3%, and its weather part has a median of 5.1% (12.6% at the 90th percentile). PM10 is the same (5.1%), and so is the all-year resampling scheme (5.2%). Network-wide, weather made 2015 a favourable year (−6.8% for PM2.5) and 2025 another (−2.9%). Weather is roughly half of a typical single-year change, so a one-year improvement says little on its own; over several years its push and pull largely cancel.

The two model families agree on average but not always on a single station’s trend: they differ by more than 5% in 355 of 822 series. That is why every ground result is shown under both. A pre-set test showed that the 2020 lockdown hardly leaks into neighbouring years under the GAM’s trend (median change 0.55%, below the 1% threshold), though most of what leaks falls on 2019 (0.95%).

Figure 3. How much does weather move annual PM2.5 numbers? Raw vs deweathered, six illustrative cities

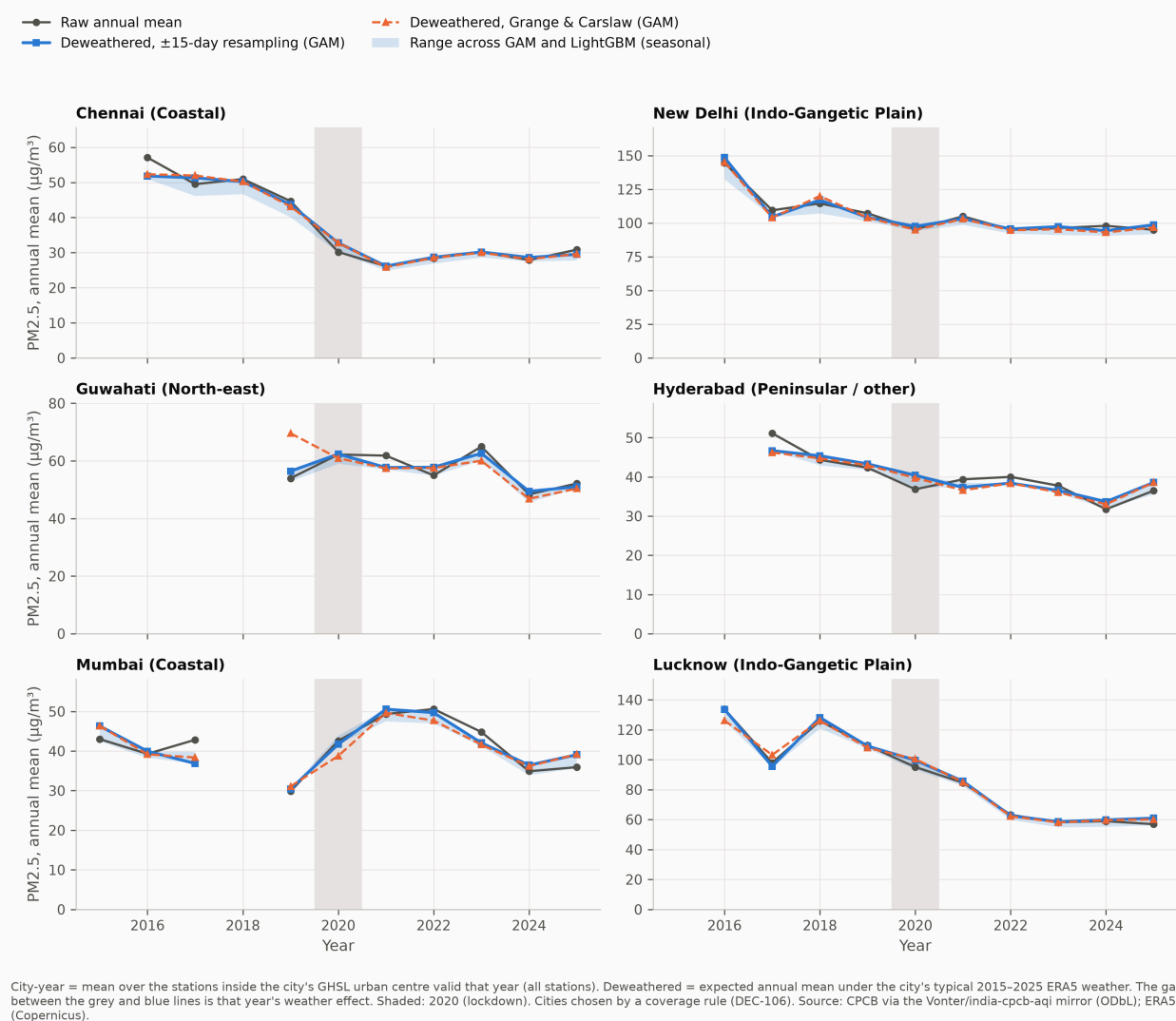


Figure 3. Raw annual mean PM2.5 (grey) and the deweathered annual mean (blue: weather resampled within ± 15 days of each date, the primary scheme; orange dashed: Grange & Carslaw all-year resampling) for six cities chosen by data coverage, one per region first. The band is the range between the two model families (GAM primary, LightGBM). Median weather part of the year-on-year change in the annual mean, per city: Chennai 5%; New Delhi 5%; Guwahati 6%; Hyderabad 4%; Mumbai 5%; Lucknow 2%; the median of these per-city medians over all 171 cities with ground data is 5.3%. 2020 is shaded: its lockdown is an emissions change, not weather, and the one-year-knot trend partly averages it out.

4.4 RQ3: did NCAP cities change differently from comparable cities?

Verdict on H1: not identified by this design (failed: (b) pre-trend Wald test). NCAP units' satellite PM2.5 did not fall relative to comparable units; the estimates point to a relative rise of about 3–5%. H1 is not identified by this design, so this is not an effect of NCAP.

The primary SDID estimate is +3.6% (95% CI +2.4% to +4.8%; +1.88 $\mu\text{g}/\text{m}^3$, +1.19 to +2.57). Rule (b) fails: the pre-listing event-study coefficients are small, but jointly different from zero ($\chi^2 = 26.8$, $p = 0.0008$). Because the satellite series is smooth, small and irregular pre-period wiggles are measured precisely enough to be “significant”; I checked on synthetic panels that the test rejects at its nominal rate when nothing is planted. Rule (c) holds (the 2016 placebo is -0.1%, -0.9% to +0.7%). Every other Layer A estimate is also an increase: Callaway and Sant’Anna +4.6%, area-weighted +3.5%, V6.GL.03 +3.2%, and the event study’s average after listing +3.1% (+1.7% to +4.5%). The HonestDiD interval stays above zero only if post-listing departures from parallel trends are at most 0.20 times the largest pre-listing one. For information only, since the verdict is already decided: the 90% interval (+2.6% to +4.6%) lies inside $\pm 5\%$, and the 95% interval excludes reductions of 20%, 30%, 40%, NCAP’s own targets, although those were set for PM10, not PM2.5.

Figure 4. Were pre-trends parallel, and when did satellite PM2.5 in listed and comparison centres diverge?

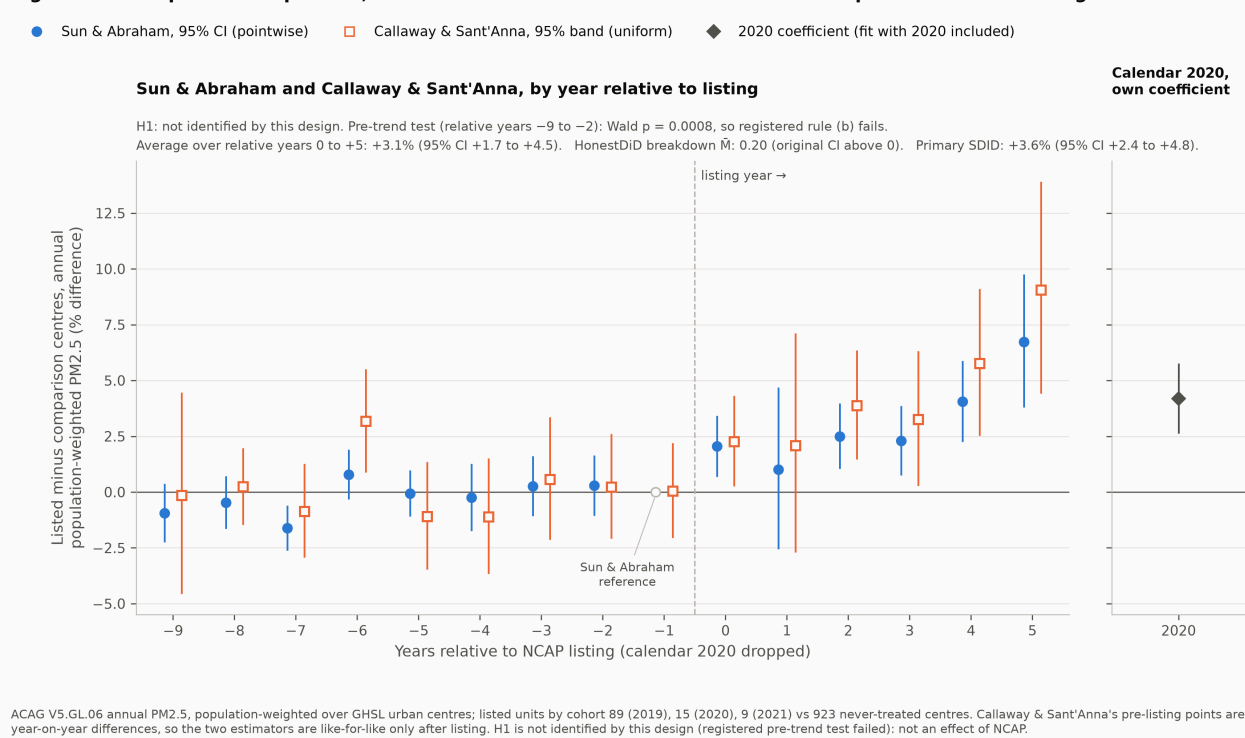


Figure 4. Event-study estimates of annual population-weighted satellite PM2.5 (ACAG V5.GL.06) in NCAP-listed urban centres minus comparison centres, by year relative to listing (calendar 2020 dropped): Sun & Abraham interaction-weighted coefficients with pointwise 95% CIs (unit and region $\times$ year effects, ERA5 covariates, SEs clustered by unit) and Callaway & Sant'Anna dynamic estimates with uniform 95% bands (never-treated controls). Listed units by cohort: 89 (2019), 15 (2020), 9 (2021); 923 never-treated centres. Pre-listing Sun & Abraham coefficients range from -1.6% to +0.8%, but are jointly different from zero (Wald $p = 0.0008$), so the registered rule (b) fails and H1 is not identified by this design. Average over relative years 0 to +5: +3.1% (95% CI +1.7 to +4.5); HonestDiD breakdown $\bar{M} = 0.20$. Right: the calendar-2020 coefficient from a fit that includes 2020 (+4.2%). Values are % differences, $100(e^{\beta} - 1)$. Satellite PM2.5 says nothing directly about PM10, NCAP's target pollutant. These are relative changes, not effects of NCAP.

Why a “relative rise” is not “pollution rose”. Satellite PM2.5 fell in both groups after 2018 (Figure S1): in NCAP units from 54.8 to 45.6 $\mu\text{g}/\text{m}^3$ (-16.8%), and in the comparison pool from 59.3 to 44.9 $\mu\text{g}/\text{m}^3$ (-24.3%). NCAP units fell less. The listed cities differ from the pool in ways the design does not fully absorb: they are much larger and less often in the Indo-Gangetic Plain, and excluding that region from both groups lowers the estimate to +1.9% (Section 5). Restricting both groups to overlapping population ranges, an exploratory check added after seeing H1, leaves the estimate where it was (+3.7% with 98 treated units; +3.0% with 42).

Figure S1. What happened to satellite PM2.5 in NCAP units and in the control pool? (descriptive)

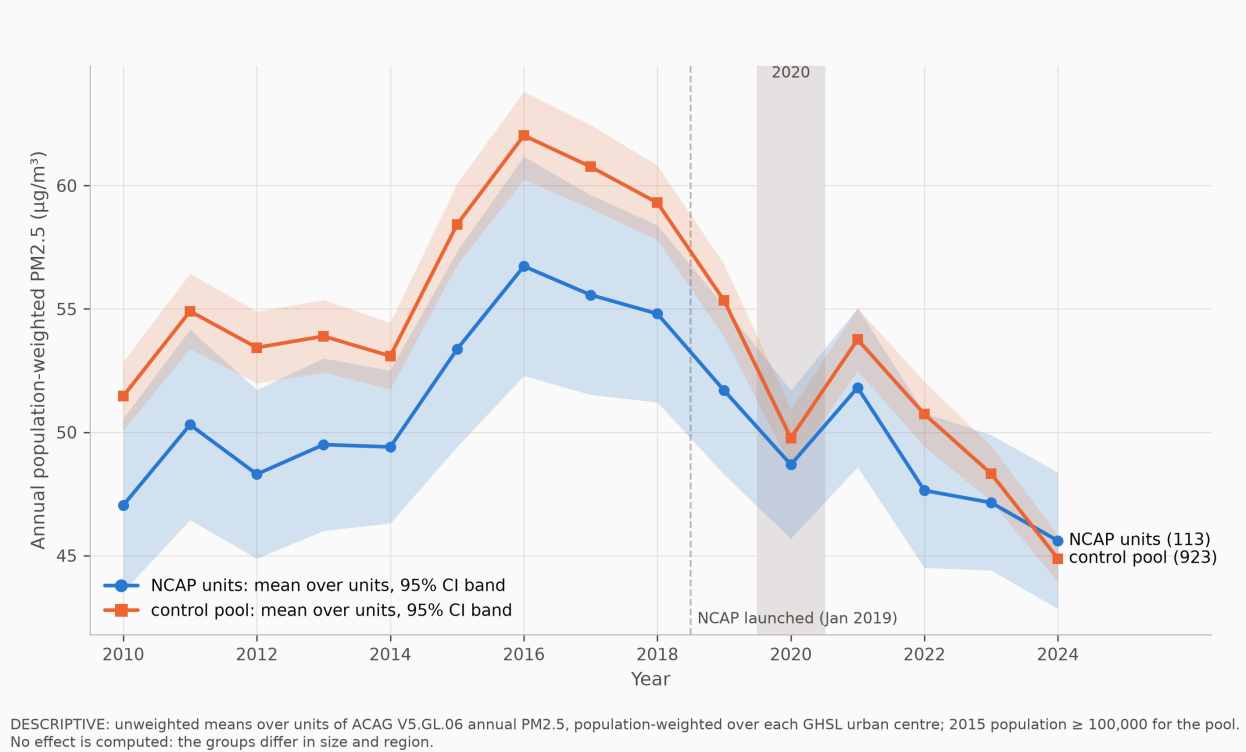


Figure S1. Mean annual population-weighted satellite PM2.5 (ACAG V5.GL.06, $\mu\text{g}/\text{m}^3$) over the 113 NCAP units and the 923 control-pool centres, 2010–2024, unweighted across units, with 95% t-intervals; 2020 shaded. From 2018 to 2024 the NCAP units’ mean went from 54.8 to 45.6 $\mu\text{g}/\text{m}^3$ (–16.8%) and the pool’s from 59.3 to 44.9 (–24.3%); both fell, the pool more. Descriptive: the groups differ in size and region, and no effect is computed from these means.

Could the satellite be echoing the new monitors? The pre-specified leakage split shows units that gained a monitor rose less (+2.5%) than units that did not (+5.5%; difference –2.8%, –5.1% to –0.5%), so the registered warning fires. This is the direction leakage would produce, because new monitors read cleaner (Section 4.2). It is a warning, not proof: the two groups also differ in size and pollution. Raw AOD tests it further (Table 5, Figure S2). On 95 treated and 741 control units with a complete AOD record, AOD’s relative change is +0.8% (–0.4% to +2.0%) against +4.2% for satellite PM2.5 on the same units: “rise not reproduced in AOD”. The gained/not-gained gap is absent from AOD: “gap absent from AOD (consistent with calibration leakage)”. These readings were written before any AOD value existed, and they have two limits.

- **The rise is not robust in AOD either way.** A non-monsoon version shows a small rise (+1.5%, +0.4% to +2.6%). In all 5 AOD specifications the AOD change is below the PM2.5 change on the same units.
- **The divergence is largest where leakage cannot act.** In an exploratory comparison added after seeing these results, the units that gained no monitor show the widest gap (+7.1% in satellite PM2.5 against –0.4% in AOD). Leakage as hypothesised would pull satellite PM2.5 down where monitors were added, so this weakens Q2 as evidence for leakage specifically. Satellite PM2.5 and raw AOD disagree about NCAP units in general, and this design cannot say why: the satellite product, or a changing link between the aerosol column and surface PM2.5.

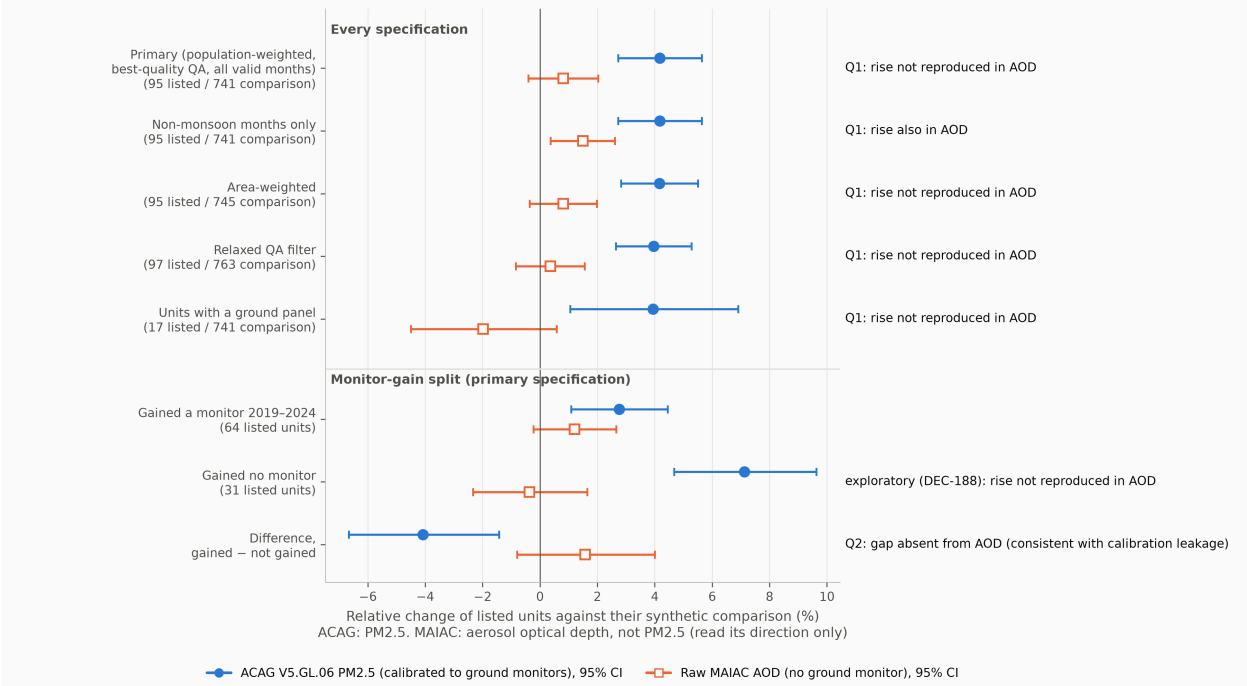
The event study on AOD also fails its pre-trend test ($p = 0.0048$), so the pre-trend problem is not only a property of the calibrated product.

Table 5. Satellite PM2.5 against raw AOD on the same units. AOD is a column measure, not surface PM2.5, so only its direction is read. Relative changes, not effects of NCAP; H1 is not identified by this design.

specification	raw AOD (direction only)	ACAG same units	PM2.5, pre-written reading
Primary: population-weighted, best-quality retrievals	+0.8% (–0.4% to +2.0%)	+4.2% (+2.7% to +5.6%)	rise not reproduced in AOD
Monitor-gain gap (gained – not gained)	+1.6% (–0.8% to +4.0%)	–4.1% (–6.7% to –1.4%)	gap absent from AOD (consistent with calibration leakage)

specification	raw AOD (direction only)	ACAG	PM2.5, pre-written reading
Exploratory: units that gained no monitor	-0.4% (-2.3% to +1.6%)	+7.1% (+4.7% to +9.6%)	diverge (exploratory, DEC-188)

Figure S2. Does a satellite signal that no ground monitor calibrates show the same relative change? ACAG PM2.5 against raw MAIAC aerosol optical depth, on the same units



Phase 8b: MODIS MAIAC MCD19A2 C6.1 AOD (0.55 μm , 1 km; NASA LP DAAC), population-weighted over the same GHSL polygons, reduced on Google Earth Engine; SDID with 500 joint-placebo replications, ACAG estimated on exactly the same units. Readings pre-specified in DEC-180; the not-gained row's reading is exploratory (DEC-188). H1 is not identified by this design (registered pre-trend test failed): not an effect of NCAP.

Figure S2. Relative change of NCAP-listed units against their synthetic comparison in raw MAIAC aerosol optical depth (squares) and in ACAG PM2.5 estimated on exactly the same units (circles), with 95% CIs. Primary: AOD +0.8% (-0.4 to +2.0), ACAG +4.2% (+2.7 to +5.6), read as “rise not reproduced in AOD”. Across the 5 specifications, AOD is below ACAG in 5 and its interval lies entirely above 0 in 1 (non-monsoon months). Monitor-gain split: AOD gained +1.2% (-0.2 to +2.7), not gained -0.4% (-2.3 to +1.6), difference +1.6% (-0.8 to +4.0); ACAG on the same units: difference -4.1% (-6.7 to -1.4), read as “gap absent from AOD (consistent with calibration leakage)”. Exploratory (DEC-188): ACAG and AOD diverge most in the units that gained no monitor (ACAG +7.1% (+4.7 to +9.6), AOD -0.4% (-2.3 to +1.6)), where calibration leakage cannot act, which weakens Q2 as evidence for leakage specifically. AOD is a column measure, not surface PM2.5, so only its direction is read. These are relative changes, not effects of NCAP: H1 is not identified by this design.

The ground layer (secondary). In the 18 PM2.5 cities (16 single-station) the deweathered before–after change is -14.0% (-22.7% to -5.3%), but against 5 control cities the DiD is -4.7% (-13.4% to +4.7%). For PM10 the ITS is -5.0% and the DiD +9.2% (-0.7% to +23.5%). With one pre-year neither can test parallel trends, and the ITS cannot separate national change from anything else.

Table 6. Layer B (ground; secondary, one pre-year), deweathered balanced panel, 95% cluster-bootstrap CIs. Not effects of NCAP.

pollutant	estimator	estimate (95% CI)	scope
PM2.5	ITS (before–after)	-14.0% (-22.7% to -5.3%)	18 NCAP cities (16 single-station) vs 5 control cities
PM2.5	ground DiD (stations)	-4.7% (-13.4% to +4.7%)	18 NCAP cities (16 single-station) vs 5 control cities
PM2.5	ground DiD (city means)	-5.1% (-15.9% to +6.7%)	18 NCAP cities (16 single-station) vs 5 control cities
PM10	ITS (before–after)	-5.0% (-11.3% to +1.7%)	13 NCAP cities (11 single-station) vs 6 control cities
PM10	ground DiD (stations)	+9.2% (-0.7% to +23.5%)	13 NCAP cities (11 single-station) vs 6 control cities
PM10	ground DiD (city means)	+12.2% (+1.3% to +26.5%)	13 NCAP cities (11 single-station) vs 6 control cities

Triangulation. Layer A restricted to the same 18 cities gives +3.8% (+1.0% to +6.6%). Against the ground DiD the pair is “uninformative”; against the ITS it is “conflict”. The registered investigation resolves the conflict at its third step: over the same cities and years, the satellite’s own before–after change is -11.1% at the stations’ cells and -11.2%

over the polygons, against –13.4% on the ground. Ground and satellite agree that these cities got cleaner. The conflict is between two questions: whether the cities got cleaner (yes), and whether they got cleaner than comparable cities (no). It is not a disagreement between two measurements.

H2 (a larger winter change) was not tested, because H1 was not supported. As exploratory numbers, winter and non-winter move alike (+2.9% and +2.8%; difference +0.1%, –1.3% to +1.4%).

4.5 RQ4: heterogeneity and mechanism

H1 is not identified, so every number in this section is a **city-level relative change**: a unit's satellite PM2.5 after listing against its own synthetic comparison. None is an effect of NCAP.

Single cities are noisy. The 113 unit estimates average +3.5% and range from –12.3% to +16.4%, but each has a standard error of about 0.049 log units. 17 have an unadjusted p-value below 0.05, and 0 of 113 survive Benjamini–Hochberg at 5%.

Pooling. All 5 hierarchical models converged at the first attempt. The between-city spread is small (0.013 log units) against each city's noise, so the estimates shrink strongly towards what the moderators predict (Figure 6). The average city-level relative change is +3.5% (+2.6% to +4.4%). Shrunk intervals lie entirely above zero for 62 units, entirely below for 1, and span zero for 50. A typical unit's 95% rank interval spans 75 of the 113 places, so the data cannot rank cities; the per-city table in the repository (docs/heterogeneity_report.md, §4) and the city explorer are alphabetical for that reason.

Figure 5. Where did satellite PM2.5 rise or fall relative to comparison units after listing?

City-level relative changes, shrunk. Not identified as effects of NCAP: the registered pre-trend test failed.

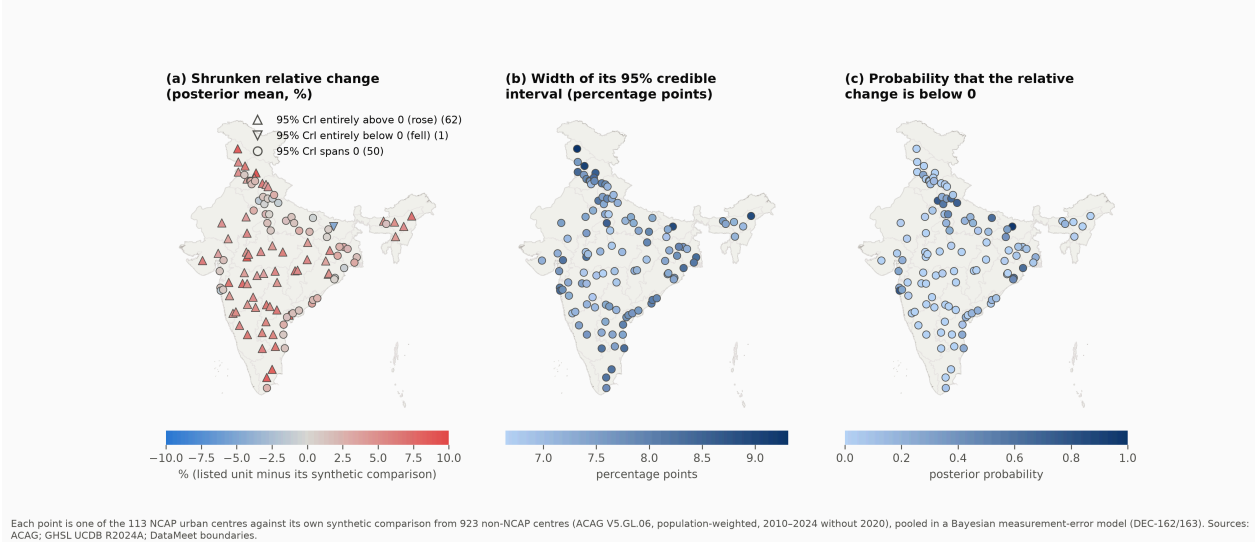


Figure 5. Maps of the 113 NCAP urban centres. (a) Each centre's shrunk city-level relative change in satellite PM2.5 after listing (posterior mean, %; blue = fell relative to its synthetic comparison, red = rose); marker shape gives its 95% credible interval: entirely above 0 for 62, entirely below 0 for 1, spanning 0 for 50. (b) The width of that interval (6.6–9.3 pp). (c) The posterior probability that the relative change is below 0. The average city-level relative change is +3.5% (95% CrI +2.6 to +4.4). Each unit is estimated against its own synthetic comparison (ACAG V5.GL.06, log population-weighted, 2010–2024 without 2020) and pooled with the five registered moderators. No city is ranked or named. These are relative changes, not effects of NCAP: H1 is not identified by this design.

Figure 6. How uncertain are city-level relative changes, and their ranks?

Purpose: to show that city ranks are too uncertain to be meaningful, so units are deliberately unnamed. Not identified as effects of NCAP: the registered pre-trend test failed.

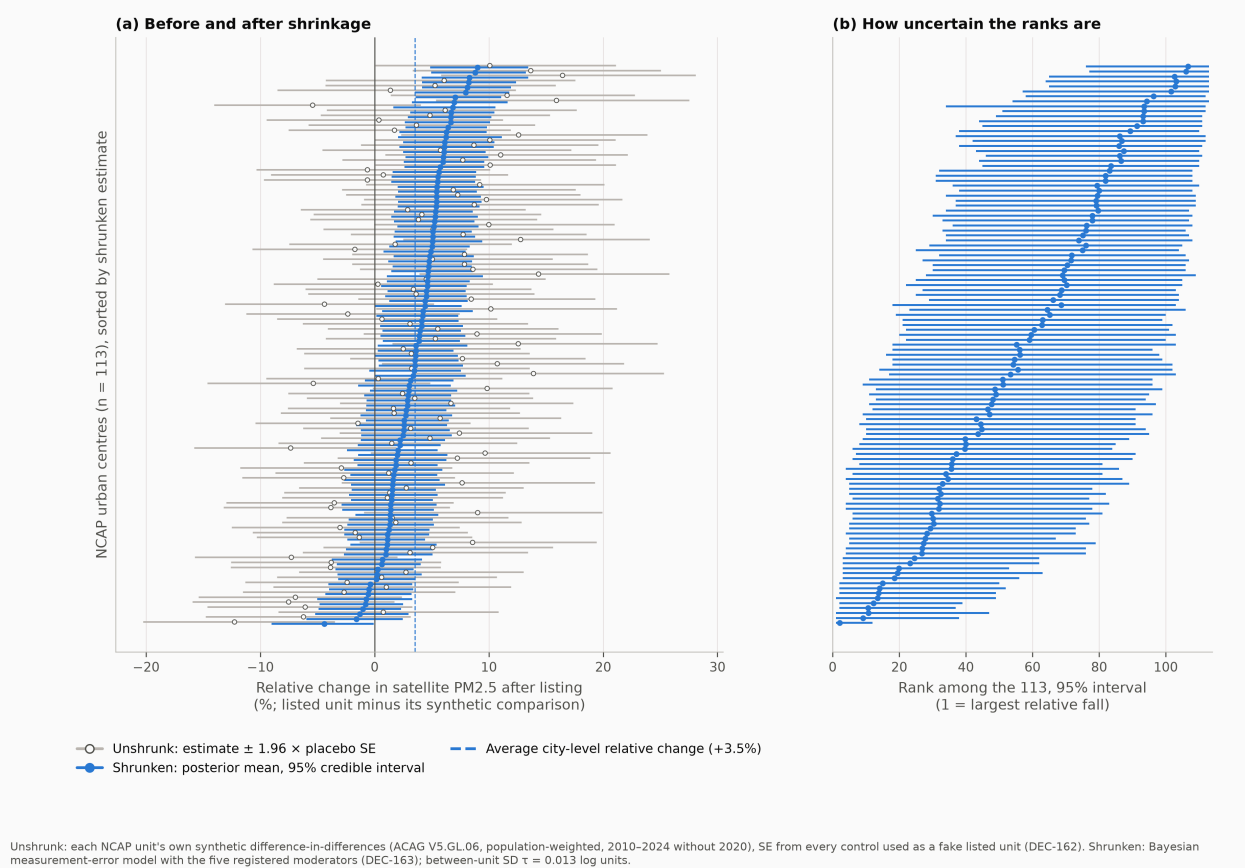


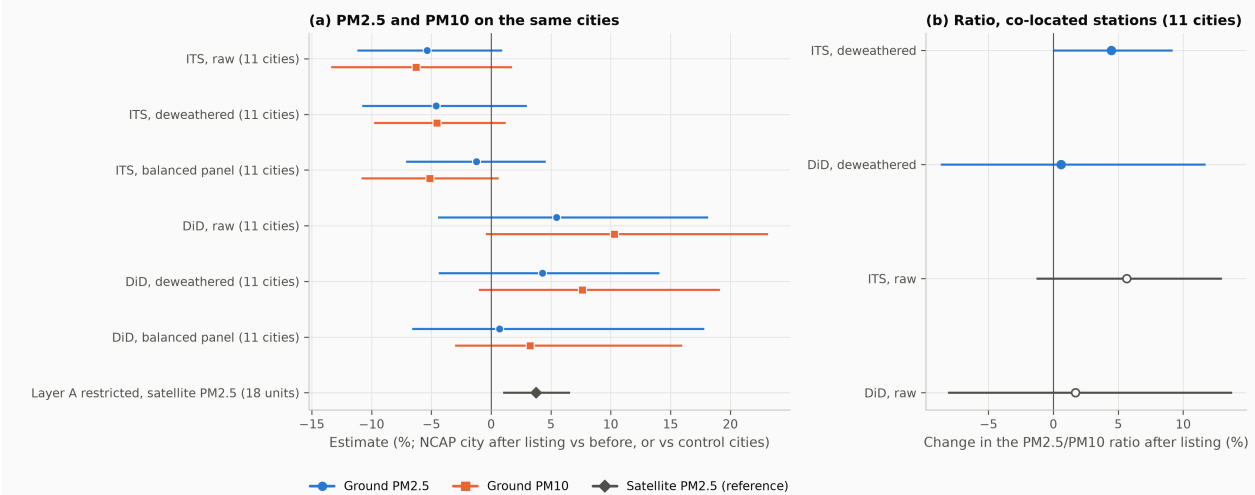
Figure 6. (a) For each of the 113 NCAP urban centres, unlabelled and sorted by the shrunken estimate: the unshrunk city-level relative change in satellite PM2.5 after listing (grey, $\pm 1.96 \times$ its placebo SE, median 0.049 log units) and the shrunken posterior mean with its 95% credible interval (blue). The between-unit SD $\tau = 0.013$ is small against the per-unit SEs, so the estimates shrink strongly towards the average city-level relative change of +3.5%. (b) The 95% interval of each unit's rank among the 113: a median width of 75 places, so the ranks are mostly uninformative. No unit is named. These are relative changes, not effects of NCAP: H1 is not identified by this design.

H5, the Indo-Gangetic Plain: the registered rule is not met. The IGP coefficient is -0.3% (-3.3% to $+2.6\%$), which spans zero. A model with the IGP alone, shown beside it and not deciding, gives -2.6% (-4.6% to -0.4%), the opposite direction to H5; the IGP correlates at 0.70 with baseline PM2.5, which the full model holds fixed. Among the exploratory moderators, coastal units show a smaller relative change (-4.0% , -6.6% to -1.3%) and XV Finance Commission units a larger one ($+3.1\%$, $+0.2\%$ to $+6.0\%$); funding channel and population correlate at 0.72, so those two share information. None of this identifies where the programme had any effect.

H3, dust control: inconclusive. Condition (iii) failed before the H3 rules were written, because the ITS pair is “conflict”. Condition (i) fails too: the ratio of PM2.5 to PM10 on co-located stations rises by $+4.5\%$ in the before–after comparison (-0.0% to $+9.2\%$, an interval that just reaches zero), and by $+0.6\%$ against control cities (-8.7% to $+11.7\%$). Condition (ii) holds for the ITS (2 of the three specifications) but not the DiD (0 of three). The scope is 11 cities with both pollutants, mostly single stations.

Figure 7. Is the ground PM10/PM2.5 pattern consistent with dust control?

H3: inconclusive. Layer B (ground): secondary, one pre-year, mostly single-station cities. Not effects of NCAP.



CPCB stations inside each urban centre, 2018 baseline, 2020 excluded; ITS = mean after listing minus before; DiD = against control-pool stations; 95% CIs: cluster bootstrap over cities (1,000). Dust control predicts PM10 falling more than PM2.5 and the ratio rising.

Figure 7. (a) Ground estimates for PM2.5 (circles) and PM10 (squares) on the 11 NCAP cities that have both, for the three registered specifications (all stations raw, all stations deweathered, balanced panel) and both Layer B estimators: interrupted time series (after listing minus before) and difference-in-differences against control-pool stations; satellite PM2.5 for the 18 Layer A units with a ground panel is shown for reference (+3.8%). (b) The change in the PM2.5/PM10 ratio on co-located stations: deweathered ITS +4.5% (95% CI -0.0 to +9.2), deweathered DiD +0.6% (-8.7 to +11.7); raw values as hollow markers. Dust control predicts PM10 falling more than PM2.5 and the ratio rising. H3 is inconclusive by its registered rule: condition (iii) already fails (the Layer A vs ITS pair is ‘conflict’), and (i) and (ii) fail too. Ground results are secondary, with one pre-year; not effects of NCAP.

Funding dose (exploratory). Across 40 million-plus units, the city-level relative change moves by -0.4% per doubling of the allocation per person (-3.5% to +2.7%); adding the IGP gives +0.1%, and dropping the highest-dose unit +0.6%. The dose is effectively “which state”: in the 11 states with two or more units, allocations per person differ by a median 0.7% within a state, while state means run from Rs 788 to Rs 3,746.

5 Robustness

Layer A. 17 of 19 registered checks agree with the primary estimate (same sign, inside its 95% interval), and every Layer A estimate is positive, from +1.4% to +4.9%. The two that do not agree are smaller but still positive: the older calibration vintage V6.GL.02.04 (+1.4%, +0.1% to +2.7%, data to 2023) and excluding the Indo-Gangetic Plain (+1.9%, +0.7% to +3.1%). Dropping any single donor moves the estimate only between +0.0349 and +0.0354 log units. Adding nearby fire activity as a covariate gives +3.1% (+1.7% to +4.5%), so nearby fire activity, as measured, does not account for it.

Table 7. Every registered Layer A check. “Agrees” = same sign as the primary and inside its 95% interval; “—” = not on the same scale or a placebo. Relative changes, not effects of NCAP.

registered check (Layer A)	estimate (95% CI)	agrees
Satellite product V6.GL.03 (rule d)	+3.2% (+1.6% to +4.8%)	yes
Satellite vintage V6.GL.02.04 (to 2023)	+1.4% (+0.1% to +2.7%)	no
Area-weighted mean (rule d)	+3.5% (+2.3% to +4.7%)	yes
Towns without a GHSL centre included as 1.87 km buffers	+3.9% (+2.7% to +5.1%)	yes
Towns included, Patancheruvu excluded	+3.9% (+2.7% to +5.0%)	yes
Asansol centre alone (without the ‘Mejia’ centre)	+3.6% (+2.4% to +4.8%)	yes
Treated units with 2015 population $\geq 100,000$ only	+3.4% (+2.2% to +4.5%)	yes
Spillover: controls ≥ 25 km from every NCAP place	+3.2% (+2.1% to +4.4%)	yes
Treatment date = first funding year	+3.1% (+1.8% to +4.3%)	yes
Anticipation: units on the 2017 list treated from 2018	+3.1% (+2.0% to +4.2%)	yes
2020 included (own per-period coefficient reported)	+3.6% (+2.5% to +4.9%)	yes
Exclude the Indo-Gangetic Plain (treated and controls)	+1.9% (+0.7% to +3.1%)	no

registered check (Layer A)	estimate (95% CI)	agrees
Patancheruvu excluded (primary)	+3.6% (+2.4% to +4.8%)	yes
2020 included: SDID per-period estimate in 2020 (own coefficient)	+4.9% (+3.1% to +6.7%)	—
Event study, 4 regions: average post-period estimate	+3.1% (+1.7% to +4.5%)	yes
Himalayan region split (event study, 5 regions): average post-period estimate	+2.8% (+1.5% to +4.2%)	yes
Callaway & Sant’Anna, never-treated controls	+4.6% (+3.0% to +6.2%)	yes
Callaway & Sant’Anna, not-yet-treated controls	+4.6% (+3.0% to +6.2%)	yes
Leave-one-out donors: the most influential donor dropped	+3.5%	yes
Placebo in time 2016 (rule c)	−0.1% (−0.9% to +0.7%)	—
Placebo in space (500 permutations)	p = 0.004	—
VIIRS fire covariate (from 2012)	+3.1% (+1.7% to +4.5%)	yes
Raw MAIAC AOD	+0.8% (−0.4% to +2.0%) (AOD, not PM2.5)	—

Layer B. 26 of 32 checks agree for PM2.5 and 26 of 32 for PM10. Those that do not are the 2019-baseline versions, which cover only the few cities listed in 2020–2021, and versions that change the model or the station set: for PM2.5 the DiD under the as-registered model and at the looser completeness rule, for PM10 both estimators under the stricter valid-hour rule. Excluding 2019, an added check, barely moves the PM2.5 ITS (−15.0%). The full table is in [docs/causal_report.md §9](#).

H4 holds in every PM2.5 version and all but two PM10 versions (Section 4.2).

What weakens the results.

- The registered pre-trend test fails, so the satellite comparison cannot be read as an effect, whatever its sign.
- The leakage warning fires, and satellite PM2.5 and raw AOD diverge. Part of the relative rise may come from the satellite product or from a changing aerosol–surface link, not from the air people breathe.
- Most ground results describe one monitor per city, and GAM and LightGBM disagree for some single cities.
- Several checks were added after I saw results: excluding 2019, dropping one station (Satna), the population-overlap SDIDs and the AOD comparison where no monitor was added. Each is labelled where it appears (Appendix A, group F).

6 Limitations

- **Identification.** NCAP cities were chosen because they were polluted or large. They differ from the comparison pool in size and region, and their pre-listing path differs from their synthetic comparison by enough to fail the registered test.
- **Satellite calibration.** The satellite product is calibrated to ground monitors, and raw AOD, which is not, shows less of the relative rise. AOD is itself a column measure, so neither is a clean reference.
- **A thin ground record.** Before 2018 the network is tiny, so the ground layer has one pre-NCAP year, cannot test parallel trends, and rests mostly on single stations.
- **The wrong pollutant for the target.** NCAP’s targets are for PM10; the confirmatory layer measures PM2.5. A PM2.5 result says nothing directly about PM10 attainment.
- **A bundle of actions.** Even an identified estimate would describe being listed, not any particular action.
- **2020.** The lockdown is dropped from the satellite panel and has its own coefficient on the ground; the GAM’s one-year trend partly averages it into 2019.
- **Weather at grid scale.** ERA5 cells miss street-level and coastal meteorology, and the deweathered trend depends on the model family for some cities.
- **Provenance.** The ground data come through a mirror of CPCB’s repository; I pinned and cross-checked it, but it is one step removed from the source.
- **Not covered.** NO2 (registered as exploratory and not carried out), January–March 2026, and the NCAP towns without a GHSL centre in the primary estimate.

7 Conclusion

In the NCAP cities I could check, reported improvements are partly an artefact of how they are measured. Where I could follow a continuous monitor from 2018 to 2025, a large part of the reported fall in PM2.5 comes from where new monitors were placed, and weather moves any single year by about half of a typical year’s change. On satellite PM2.5,

NCAP cities improved, but not by more than comparable cities; my registered design cannot say whether NCAP made any difference, because its own pre-trend test failed, and I report that as the result rather than the estimate.

Five recommendations follow from these findings, each within what the data can support.

1. **Report a fixed panel of stations beside the all-station average**, so that network growth does not show up as improvement.
2. **Report weather-normalised values beside raw ones**, and judge progress over several years, not one.
3. **Publish station start dates, moves and instrument changes**, so that composition and level shifts can be checked.
4. **Report PM_{2.5} as well as PM₁₀**, since health effects and the satellite record are in PM_{2.5}.
5. **Evaluate against comparison cities with a long pre-period, with the analysis registered in advance**, so that a null or unidentified result is as informative as a positive one.

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- NASA FIRMS. I acknowledge the use of data from the NASA LANCE FIRMS (<https://earthdata.nasa.gov/firms>), part of the NASA Earth Science Data and Information System (ESDIS).
- DataMeet India community. India boundaries. <https://github.com/datameet/maps>. CC BY 4.0.
- GeoNames. <https://www.geonames.org>. CC BY 4.0.
- Natural Earth. 1:10m coastline. <https://www.naturalearthdata.com>. Public domain.

Appendices

Appendix A. Deviations from the registered plan

I registered my analysis plan on OSF on 27 September 2026 (<https://osf.io/jksne/>). The registered file is [docs/analysis_plan.md](#) at commit [6e24eca](#), and the code gate that allowed post-2019 estimates was opened in a separate commit, [0d9aa42](#), citing it. On 1 October 2026 I posted a clarification of how I read some of the registered decision rules. The registration promised that every deviation would be dated, justified and listed in this report. This appendix is that list.

Every entry gives the date I decided, the commit that first recorded the decision in the public repository, its entry in [docs/DECISIONS.md](#), my reason, and whether I made it before or after I had seen the estimate it affects. Git shows when a rule was committed, not when I first looked at a number; where the order of events matters, the DECISIONS entry states it.

The entries fall into six groups:

- **A.** Deviations from the registered plan.
- **B.** Registered analyses that I did not carry out, or that the data could not support.
- **C.** Clarifications of registered rules, posted on OSF on 1 October 2026.
- **D.** Departures from rules I fixed after registration, and from the original proposal.
- **E.** Places where the registered text was incomplete or could not be applied literally, and how I read it.
- **F.** Analyses I added after registration. Each is labelled as such wherever it appears.

Implementation fixes that changed no specification (for example DEC-083, DEC-152 and DEC-158) are not listed.

A. Deviations from the registered plan

The registered text fixes the rule that chooses the deweathering model family, the data-validity rules and the exclusion rules. It does not spell out the model specification or the weather-resampling scheme. I list A1 and A3 as deviations anyway, because each changes what the registered analysis would have computed. Wherever a result depends on A1–A3, I also report the version with none of them (“as registered, no deviations”: the original GAM trend, Grange & Carslaw resampling, registered flags only).

#	Registered	What I did instead	Decided	DEC	Why	Timing
A1	Deweathering by the proposal’s method (Grange & Carslaw 2019), whose default resamples each day’s weather from any time of year.	Each day’s weather is re-sampled from ERA5 days within ± 15 days of the same date. The all-year scheme is run on every series as a sensitivity analysis.	2026-09-27, 083370b	DEC-109	All-year resampling pairs a date with weather that never occurs in that season, so the model predicts outside the conditions it was fitted on.	After the pilot’s deweathering output on 20 stations; before the full deweathering run, before H4 and before any treatment estimate.

#	Registered	What I did instead	Decided	DEC	Why	Timing
A2	Data are excluded only by the registered flags, completeness rules and control-pool rules.	A near-constant-analyser rule removes station-years in which an instrument barely moved for a month or more while its neighbours did. The registered flags alone are the sensitivity analysis.	2026-09-27, 083370b	DEC-110	The pilot found a stuck instrument that the registered flatline rule cannot catch, because its hourly values still change slightly.	After the pilot; before the full deweathering run, before H4 and before any treatment estimate.
A3	The family-selection rule (better median out-of-sample R^2 under blocked, forward-chaining CV) chooses between LightGBM and the GAM.	The GAM's trend has knots exactly one year apart, so it cannot follow seasons or weather episodes. The original GAM is kept as a sensitivity family, and the registered rule was re-applied to the refitted GAM (DEC-117), which it chose again (DEC-118).	2026-09-28, 72f2fc3 (pushed before the refit)	DEC-116	The original trend absorbed part of the seasonal cycle and year-specific weather, and under all-year resampling it produced physically impossible values.	After the full deweathering run and after the registered rule had chosen the original GAM (DEC-114, DEC-115); before H4 and before any treatment estimate.

B. Registered analyses not carried out or not computable

#	Registered	What happened	Decided	DEC	Why	Timing
B1	A FY2025-26 sensitivity analysis for ground results, adding January–March 2026.	Not computable.	2026-09-30 (H4), 2026-10-01 (Layer B)	DEC-128, DEC-149	The deweathering models are fitted on days up to 31 December 2025 (DEC-101), because January–March 2026 exists only in OpenAQ's unvalidated real-time feed (DEC-079). So no deweathered values exist for 2026.	Known before H4 and Layer B were computed.
B2	Completeness 90% as a sensitivity analysis for ground results.	Not computable for H4 or Layer B.	2026-09-30	DEC-132	At 90% completeness no NCAP city has a station valid every year from 2018, so there is no balanced panel.	Found when H4 was first computed.
B3	NO ₂ as a secondary pollutant (registered as exploratory).	Not carried out.	2026-10-03	DEC-208	I found this omission while compiling this list. NO ₂ was ingested and cleaned but never analysed.	Found after every other result had been seen.
B4	H2 (a larger winter reduction), tested only if H1 is supported.	Not tested, as registered; the winter and non-winter estimates are reported as exploratory.	2026-10-01	DEC-151	H1 was not supported.	Follows the registered sequence.

The registered “if time allows” check on raw MAIAC aerosol optical depth was carried out (Phase 8b; its rules are E12), and so was the VIIRS fire-covariate check (E11).

C. Clarifications of registered rules

These change no hypothesis, threshold or estimator. Some registered rules call effects “changes” that can be read with either sign; I fixed one reading for each. On 1 October 2026 I posted a summary of these readings on the OSF registration (saved as [docs/osf/clarification_2026-10-01.md](#)). The full set of readings is DEC-135, so C2 covers more than the posted text.

#	Registered text	My reading	Decided	DEC	Why	Timing
C1	H4: “raw all-station change minus deweathered balanced-panel change ... supported if the mean across cities is positive”.	“Change” means improvement: H4 = reported fall minus corrected fall, positive when the reported number overstates the improvement.	2026-09-30, 29874cf	DEC-127	Read literally on concentration changes (a fall is negative), the rule would call the opposite of the hypothesis “supported”. The hypothesis text fixes	Committed before H4 was first computed (132b788, the same morning). The commit was pushed publicly on 1 October 2026, af-

#	Registered text	My reading	Decided	DEC	Why	Timing
					which sign counts as support.	ter H4 had been computed, as the OSF clarification states.
C2	Every decision rule in plan §5.	One sign convention for all effects (negative = a reduction); H1 rule (d) as “point estimate < 0”; the equivalence bounds kept exactly as registered (ln 0.95 to ln 1.05); H2 as a negative winter-minus-non-winter difference; H3 as a larger PM10 reduction with PM10 actually falling; H5’s coding; “agrees” judged on the same scale as the primary; a fixed order for the layer-disagreement categories, which overlap, with a fifth “unclassified” case reported under that name; Benjamini-Hochberg on two-sided p-values.	2026-09-30, ba397a2	DEC-135	After C1 I audited every registered rule for the same kind of ambiguity.	Committed and pushed before any Phase 7 estimate (the first is 655fb65 , 2026-10-01).

D. Departures from rules fixed after registration, and from the proposal

#	Earlier rule	What I did instead	Decided	DEC	Why	Timing
D1	DEC-103: in cross-validation, the trend for an unseen test year is clamped at the last training day.	A test day takes the trend of the same calendar day one year earlier. Both conventions are reported.	2026-09-27, d57c495	DEC-107	Clamping carried the last training day’s season (late December) into the whole test year. This changes how the registered selection metric is computed, not the rule itself.	After seeing the GAM’s pilot results and before LightGBM’s; before the full run. The full run’s family choice depended on the convention (DEC-114); after A3 the GAM is chosen under both (DEC-118).
D2	DEC-103: the number of weather draws is the smallest that meets a convergence rule (300 in the pilot).	500 draws for every series.	2026-09-27, 083370b	DEC-113	300 met the rule only narrowly; 500 is inside the proposal’s range of 500–1,000.	After the pilot; before the full run. Unrelated to any treatment comparison.
D3	DEC-130: the GAM–LightGBM disagreement diagnostic splits each model’s weather effect on the log scale.	The split is made on the annual-mean scale, into modelled weather and model misfit. The failed version stays in the outputs.	2026-09-30, 132b788	DEC-133	The log-scale split failed its own pre-set additivity check. The diagnostic chooses nothing; the registered rule chose the family.	After the check failed; it affects no hypothesis test.
D4	DEC-150 (and the proposal’s figure 1): a final “policy” step drawn from the corrected ground change, followed by a “remaining change” bar.	Figure 1 has no policy step and no remaining-change bar. The satellite relative change is drawn on its own axis, labelled “not identified as an effect of NCAP”.	2026-10-03, 09209e7 ; approved 406f544	DEC-191, DEC-197	After H1 was not identified, subtracting the satellite estimate presupposes the attribution the design failed to establish. The two quantities also differ in time base and definition. No number changes.	After the H1 verdict (DEC-151).
D5	No rule existed for which cities illustrate figure 1.	Panel (c) shows four cities chosen by a rule on station counts: in each region present, the city with the most stations valid in 2025, then the next city with the most, drawn alphabetically. Every city is in figure S3.	2026-10-03, 09209e7 ; approved 406f544	DEC-191, DEC-197	Composition can matter only where the network changed, and station counts use no outcome. The figure’s panel heading and caption say when the rule was written.	After I had seen every city’s decomposition (docs/composition_report.md §1d). The rule uses station counts only, but the choice of rule was not blind.

#	Earlier rule	What I did instead	Decided	DEC	Why	Timing
D6	The proposal's figure questions, e.g. figure 5: "Where did NCAP work, and where not?"	Questions that presuppose an effect are reworded, e.g. "Where did satellite PM2.5 rise or fall relative to comparison units after listing?"	2026-10-03, 09209e7	DEC-190	The same reason as D4.	After the H1 verdict.
D7	DEC-167: figure 7 shows the satellite PM2.5 reference in panel (b).	The reference row is in panel (a).	2026-10-02; logged 2026-10-03, 4eb69f7	DEC-170	Panel (b)'s axis is the change in the PM2.5/PM10 ratio, so a PM2.5 estimate drawn there would read as a ratio. Layout only.	After the figure was drawn; no number changes.
D8	DEC-179: every MAIAC sensitivity specification is estimated with the monitor-gain split.	The sensitivity specifications estimate the overall relative change only and are classified by Q1 only; the split (and Q2) is estimated on the primary specification.	2026-10-03, 05ab2f7	DEC-182	The split on every sensitivity would roughly triple the run time for numbers that classify nothing.	Before any AOD value existed.
D9	DEC-178: export to Earth Engine assets, and stop if the pilot projects more than 100 EECU-hours of compute.	Export to a private Google Drive folder, with md5 checks; the compute limit was raised to 400 EECU-hours after I moved the project to the Contributor tier.	2026-10-03, d6289c5 , b7eca89 , 9ae23d9	DEC-183, DEC-184, DEC-185	The project had no asset area to write to, and the pilot tripped the stop rule. Neither change alters any computed value.	Before the export, so before any AOD value existed.
D10	The proposal's dashboard: each city's raw, deweathered, composition-corrected and synthetic-comparison trends.	The dashboard leaves out the synthetic comparison's trajectory.	2026-10-03, a139fe3	DEC-203	A drawn gap between a city and its synthetic comparison invites a causal reading that the design did not identify; the per-unit trajectories were also not saved.	After the H1 verdict.

Earlier departures from the proposal (satellite product versions, the ground-data source, dropping WorldPop and pyGAM, VIIRS instead of MODIS fire data, the data window) were made before registration and are part of the registered plan; they are described in Section 2 and logged in DEC-001 to DEC-046.

E. Readings of incomplete or inapplicable registered text

Each of these fills a gap the registered text leaves open, or resolves a case where it cannot be applied literally. Each was committed before the estimate it governs. E10 to E12 were written after the Phase 7 estimates had been seen, which is why I list them here with their timing.

#	Registered text	My reading	Decided	DEC	Timing
E1	H4 "per NCAP city with ground data".	Per GHSL urban-centre unit: NCAP cities sharing one polygon form one unit, so each ground series has a matching satellite unit.	2026-09-30, 29874cf	DEC-125	Before H4.
E2	H4 names no pollutant.	PM2.5 and PM10 are tested separately and never pooled; the all-station set is limited to stations with deweathered series.	2026-09-30, 2254a70	DEC-131	Before H4.
E3	Placebo in space, 500 permutations.	The permutation p-value of the primary estimate against the same 500 joint-placebo replications that give its standard error.	2026-10-01, eebaecc	DEC-139	Before any Phase 7 estimate.
E4	Event study with relative years -9 to +5 and -1 omitted; 2020 dropped from the primary panel.	For the 2021 cohort, relative year -1 is calendar 2020, which is dropped, so its reference is -2. Relative years outside -9 to +5 get their own indicators rather than being binned.	2026-10-01, eebaecc	DEC-142	Before any Phase 7 estimate.
E5	Callaway & Sant'Anna, doubly robust.	No covariates are added, because the registered text names none; without covari-	2026-10-01, eebaecc	DEC-144	Before any Phase 7 estimate.

#	Registered text	My reading	Decided	DEC	Timing
		ates the estimator is the unconditional difference-in-differences.			
E6	Sensitivities “exclude Patancheruvu” and “Himalayan region split”.	Patancheruvu has no unit of its own in the primary (it lies inside Hyderabad’s centre), so its exclusion is run on the buffered-towns version. The Himalayan split changes only region × year effects, so it applies to the event study, not to SDID.	2026-10-01, eebaecc	DEC-147	Before any Phase 7 estimate.
E7	Layer B: “deweathered interrupted time series” and “NCAP vs non-NCAP stations with year effects”.	With one pre-year, the ITS is the before–after contrast of mean log levels. The ground DiD is estimated per listing cohort against never-treated control stations, with a cluster bootstrap stratified by cohort and control.	2026-10-01, eebaecc	DEC-148	Before any Layer B estimate.
E8	“Layer A restricted to the units that have Layer B stations”; H3’s “Layer A is not in the opposite direction”.	Two pairs are classified (against the ground DiD and against the ITS); H3 condition (iii) holds only if neither pair is “conflict”.	2026-10-01, eebaecc	DEC-150	Before any Phase 7 estimate.
E9	Layer B at the 2019 balanced-panel baseline (registered sensitivity).	At that baseline the 2019 cohort has no pre-year, so it contributes no contrast; those rows cover only the cities listed in 2020–2021, and say so.	2026-10-01, 9a04145	DEC-157	Found when the first run stopped with an error, before any Layer B number was written.
E10	The hierarchical model of city effects, H5 and H3 (plan §5).	One SDID per treated unit with placebo-based standard errors; the moderators’ coding (reference region = peninsular/other plus north-east), priors, sampler and convergence rule; H3’s three specifications mapped to the Layer B series, with condition (ii) required for both estimators; the units that receive a funding dose.	2026-10-02, 2f80dc4	DEC-162 DEC-166	After the Phase 7 estimates had been seen; before any Phase 8 estimate. H3’s verdict was already fixed by condition (iii), which had failed.
E11	“VIIRS fire covariate (from 2012)”.	Log fire radiative power within 100 km of each unit, added to the event study; the check is judged by the “agrees” rule.	2026-10-03, 4eb69f7	DEC-171	After the primary estimate had been seen; before any fire value was read.
E12	Raw MALAC aerosol optical depth “if time allows”, with no further detail.	The product, QA filter, aggregation, coverage rules, design and a pre-written reading of every possible outcome (two questions, read for direction only).	2026-10-03, 85acc2a	DEC-174 DEC-181	After the primary estimate had been seen; before any AOD value existed.

F. Analyses added after registration

None of these is a decision rule, and none can change a registered verdict.

#	What I added	Label in the outputs	Decided	DEC	Timing
F1	A pre-set test of whether the 2020 lockdown smears into 2019 and 2021 under the one-year-knot trend, with a rule to add a lockdown term if it did. The rule was not triggered.	Reported in the deweathering report	2026-09-28, f09c22e	DEC-119, DEC-122	Rule committed before the test was fitted.
F2	Layer B re-estimated without 2019.	“added 2026-09-28”	2026-09-28, 7ecd0a1	DEC-123	After the lockdown test (F1); before any Layer B estimate.
F3	One station (Satna, site_1433) dropped from every registered-flags version.	“added after inspecting the data”	2026-09-28, 7ecd0a1	DEC-124	After inspecting that station’s data; before H4 and Layer B.
F4	H4’s weather part split into modelled weather and unmodelled change. H4’s value and test are unchanged.	Described as an added decomposition	2026-10-01, 86e62c7	DEC-136	After H4 had been seen; before the split was computed.
F5	The ground DiD on city means beside the station-level DiD, and a LightGBM	“added before computing”	2026-10-01, eebaecc	DEC-148, DEC-149	Before any Layer B estimate.

#	What I added	Label in the outputs	Decided	DEC	Timing
	follow-up at the 2019 base-line.				
F6	Satellite SDID restricted to overlapping ranges of city population (two variants).	“exploratory, added after seeing H1”	2026-10-01, a90213d	DEC-155, DEC-161	After the H1 verdict. The second variant was written after seeing the first variant’s population balance, and before either was run.
F7	Satellite PM2.5 against raw AOD in the units that gained no monitor.	“exploratory”	2026-10-03, d43cddb	DEC-188	After the Phase 8b results had been seen; both group estimates were already printed.

Appendix B. Reproducibility

Every processed file, figure and number in this report is rebuilt from the raw downloads by one Snakemake workflow. The repository README gives the commands.

- **Raw data are immutable.** Each source’s `data/raw/<source>/MANIFEST.csv` records the URL, upstream identifier, download date and SHA-256 checksum of every file. The raw files themselves are not in the repository; the downloaders fetch them again, and a file whose checksum differs is refused. Re-downloading needs free accounts for the Copernicus Climate Data Store, OpenAQ and NASA FIRMS, and a registered Google Earth Engine project for the aerosol optical depth tables.
- **Pinned environments.** The analysis environment (Python and R) and the separate report and site environment (Quarto only) are each locked with `conda-lock` for Windows and Linux. R packages not on conda-forge come from a dated CRAN snapshot or a pinned commit.
- **The pre-registration gate.** Every step that compares NCAP with non-NCAP units after 2018 checks `config/gate.yaml`, which names the registered plan’s commit.
- **No typed numbers.** Every number in this report, the results summary and the policy brief is read from a file that `python -m src.report.values` writes from the pipeline’s outputs, and a test fails if a result-like number appears in the text by hand. The same test runs a wording check that refuses any phrase that attributes a result to NCAP or ranks cities.
- **What is not byte-identical.** Figures rebuild byte for byte. Some tables and spatial files differ in byte order or embedded timestamps between runs but not in content. Earth Engine re-exports agree to about eight significant figures, so the aerosol tables are pinned by checksum.
- **A rebuild from a clean clone** of the public repository, with fresh environments and the raw data checked against every manifest, is recorded in `docs/DECISIONS.md` (Phase 10).

Appendix C. Licences and attributions

- **Code** (the pipeline, tests and workflow): MIT licence.
- **Text and figures** (this report, the summary, the brief, the figures and the documentation): Creative Commons Attribution 4.0 (CC BY 4.0).
- **Ground-derived data** (any table derived from the CPCB station data): Open Database License 1.0 (ODbL), as the source’s licence requires.

Notice for figures and tables that use ground data. Contains information from `india-cpcb-aqi`, a mirror of CPCB’s data repository, which is made available under the ODbL 1.0; some contents © CPCB. The adapted station data behind these works, and the method that produced them, are available under the ODbL 1.0 in the public repository.

Other sources (full citations in the references):

- ACAG satellite PM2.5, Washington University in St. Louis (CC BY 4.0).
- Contains modified Copernicus Climate Change Service information (2026): ERA5 (CC BY 4.0).
- GHSL Urban Centre Database and GHS-POP, © European Union 1995–2026 (CC BY 4.0); I joined polygons into units and computed population-weighted means.
- India boundaries by the DataMeet India community (CC BY 4.0).
- GeoNames (CC BY 4.0); Natural Earth (public domain).
- NASA LANCE FIRMS and MODIS MAIAC (NASA LP DAAC), NASA open data.
- OpenAQ, and the original data providers.

Appendix D. Supplementary figures

Figure S3. City by city: reported change, the parts removed, and the corrected change (alphabetical)

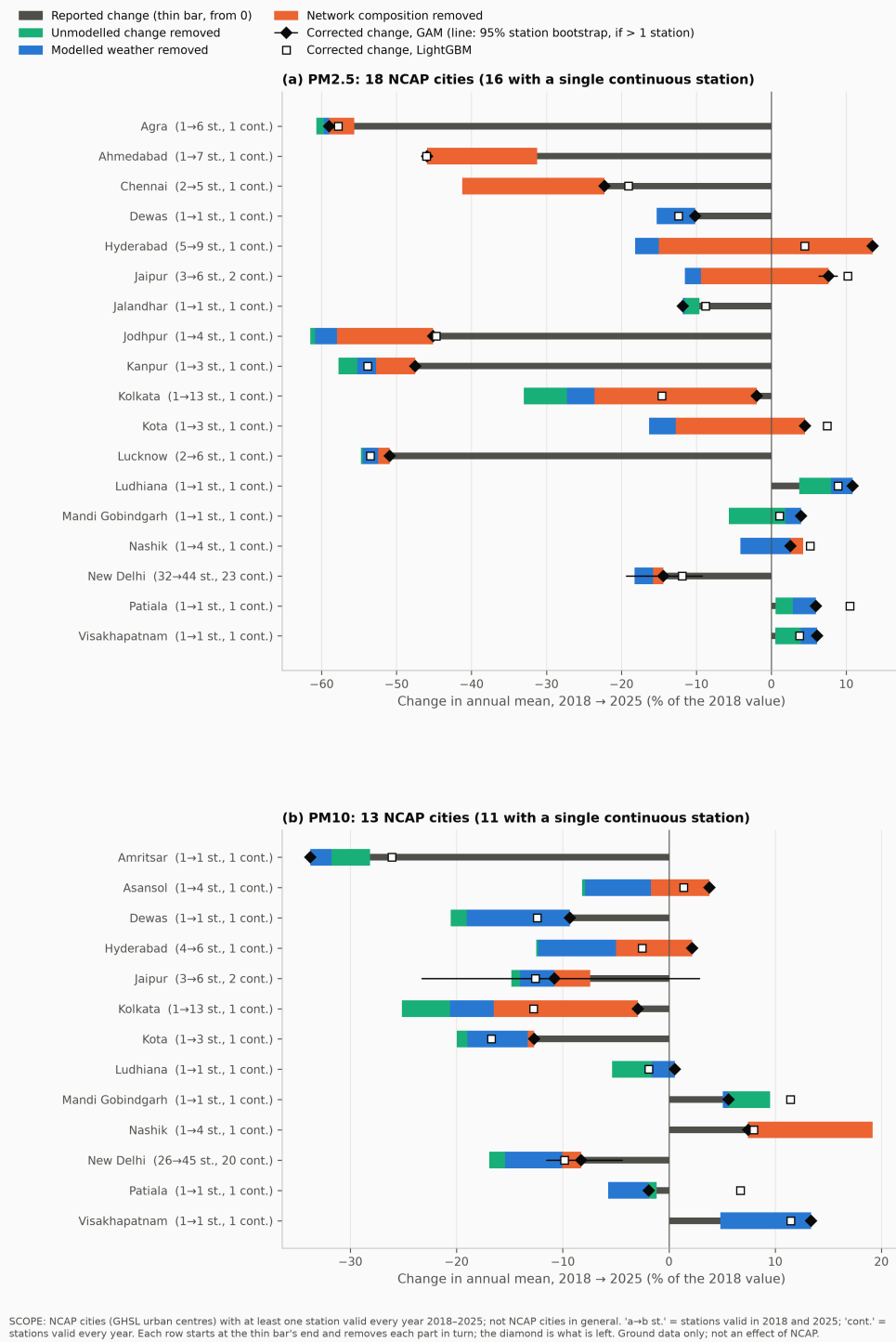


Figure S3. Every NCAP city in H4's scope, listed alphabetically (never ranked): (a) PM2.5, (b) PM10. Each row shows the reported change in the annual mean 2018 → 2025 (thin bar), then removes the unmodelled change, modelled weather and network composition; the diamond is the weather- and composition-corrected change (GAM; line: 95% station bootstrap where the city has more than one station), the square the same with LightGBM. Most cities have one continuous station, so their corrected change is one monitor's record. Ground data only; nothing here is an effect of NCAP.

Figure S4. How much of NCAP cities' reported fall in PM10 survives weather and network correction?

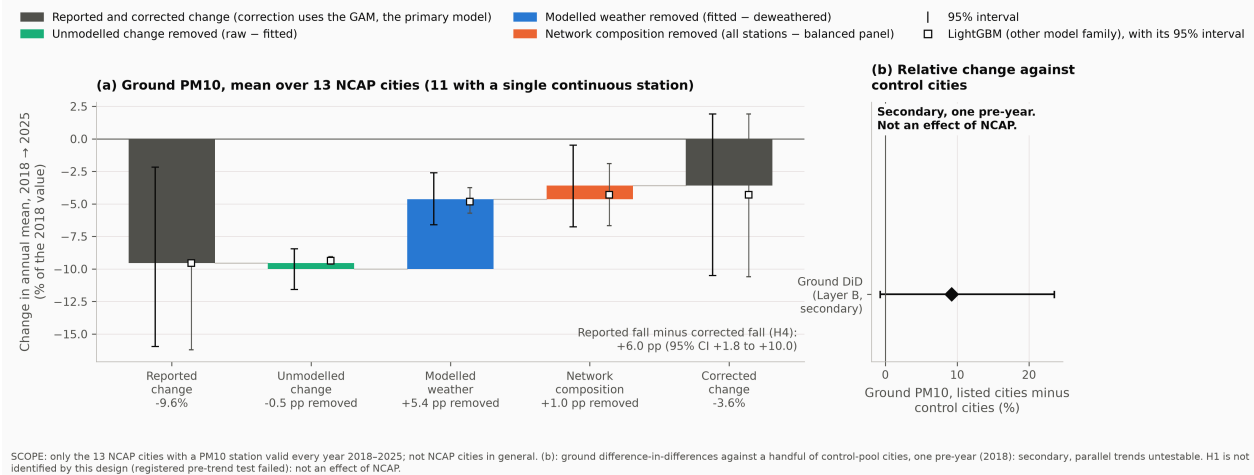


Figure S4. (a) Mean change in annual ground PM10 from 2018 to 2025 over the 13 NCAP cities with a PM10 station valid every year (11 with one such station): reported –9.6%; unmodelled change –0.5 pp removed, modelled weather +5.4 pp removed, network composition +1.0 pp removed; corrected –3.6%. H4 = +6.0 pp (95% CI +1.8 to +10.0). GAM bars, LightGBM squares, 95% cluster-bootstrap intervals. (b) Ground difference-in-differences against control-pool cities (Layer B, secondary, one pre-year): +9.2% (95% CI –0.7 to +23.5). There is no satellite PM10. H1 is not identified by this design (registered pre-trend test failed): not an effect of NCAP.